The stochastic structure of Hong Kong racing performance

An exploratory latent-performance study, and what of it belongs in a betting model

Alternative Performance-Distribution Research Project

31 August 2026

What this document is. An exploratory study of the distribution of horse performance in HKJC Class 1–5 racing, 2008–2026, built from individual finishing times and per-section sectional times. It is a companion to an existing production winner-probability model, which it treats as read-only and as the reference for what a useful result would have to look like.
What it does not establish. Nothing here is a profitability claim. Every market price used is a result-page price, recorded after the pool closed, at which no bet could have been placed; no dividends were loaded and no return is computed anywhere in this project. Every season in the archive has been development data, here and in the reference project, so no window is genuinely unseen. The confirmation step this study cannot substitute for is a prospective test of a locked model against timestamped pre-race price snapshots.

Contents

1 Executive summary

The reference production model beats the market by -0.0045 of race log loss. That number is the yardstick for everything below: a piece of performance structure is worth having only if it moves a meaningful fraction of it.

Fourteen structural questions were asked. The answers divide unevenly, and the division is the result.

What is real, and useful

- **Sections are exactly comparable across distances.** HKJC times every race in 400 m sections measured back from the finish, the first carrying the remainder — verified on all 13,309 races in the population. So "the last 400 m" is one physical quantity in a 1200 m sprint and a 2200 m staying race. Both of the reference project's own sectional engines read the first and last section only; the middle of the race was unexamined, and this study examines it.
- **Jockey ability is identifiable, and is a far better forward predictor than horse ability.** Jockey identity alone forecasts relative performance at out-of-sample R^2 **0.048** against horse identity's 0.012, and at a forward scale of **1.05** against the horse effect's 0.39. Not because it is larger — it is a fifth the size — but because it is far better determined: about 1,000 rides per jockey against 15 starts per horse, and no drift. Two independent estimators agree at Spearman 0.970.
- **Horse ability drifts with a half-life of about two months.** Forward correlation is monotone in recency and peaks at 60 days (0.150 against a static 0.114). A static ability estimate is over-scaled by a factor of 2.7.
- **Horse-specific volatility is real and forecastable.** 61.8% of the observed spread in per-horse log residual standard deviation is signal rather than sampling noise; a horse one standard deviation more volatile has 1.43× the spread, and the property persists out of sample at $z \approx 8$.
- **The bad tail is heavier, and is partly a recorded event.** The out-of-sample residual has skewness -0.729 and excess kurtosis $+0.810$. Of runs more than three standard deviations below expectation, **46%** carry a severe stewards' phrase against a 9.5% base rate — a lift of 4.8, monotone across all seven residual bands.
- **Race-level shared shocks are large, and track speed is a property of the day.** The shared shock is 32% of absolute residual variance, about half a second at 1200 m. A running mean of earlier races on the same card correlates **0.50** with a race's own shock, rising to 0.58 by the eighth race of a meeting.

What is real, and not useful

- **Horse × environment sensitivity exists only for some variables.** Venue (Happy Valley against Sha Tin) and surface sensitivity persist out of sample at $z = 8.3$ and $z = 6.4$. Going sensitivity does **not** ($z = 1.37$) — even though going moves absolute race time by 1.2 s. This is precisely the outcome the research plan nominated in advance as its most valuable possible finding, and it lands.
- **The sectional profile is more persistent than final time, for the wrong reason.** Fourteen sectional quantities beat the best final-time target on persistence, the strongest

reaching forward $r = 0.483$ against 0.278. But persistence and informativeness are nearly *inversely* ordered: the same quantity predicts its own next value at 0.483 and next *performance* at 0.045. What persists is running *style*, a habit; style persists precisely because it carries little ability information.

- **The market underprices lightly-raced horses, and the production model already knows.** A measure of ability uncertainty essentially uncorrelated with the price (-0.004) predicts the outcome given the price at $t = +15.0$. But the incumbent's own 28 columns already reconstruct **73%** of it. There is nothing left to add.

What is not there

- **Horse × jockey compatibility is not forecastable.** The median pair has **one** ride and 87.8% have fewer than five. On pairs with enough rides to test, the early-to-late correlation is indistinguishable from zero (z from -0.4 to $+0.8$) on a design that would have shown $r = +0.21$ for a true pair standard deviation of 0.20. The *in-sample* fit meanwhile reports a 10.3% variance share — a clean illustration of why training fit is not evidence.
- **The draw effect is large and fully priced.** Relative performance falls 0.25 standard deviations from the inside to the outside draw; the market residual across the same bands is non-monotone and within 0.2 percentage points of zero, and a date-atomic draw-bias estimate predicts the market's error at $r = 0.007$.
- **No elaboration of the shock distribution improves the probabilities.** Once a single utility scale is fitted per variant, plain homoskedastic normal shocks are the best of six nested generative variants — heteroskedasticity, horse-specific volatility, Student-t tails, latent-state uncertainty and a shared pace factor all fail to improve on it. This independently reproduces the reference project's own probit result that the utility scale is the whole game, now with measured heteroskedasticity and a sectional-derived pace loading rather than a position habit.

Groups of horses, rather than individuals

Pursued after a first reading of the individual-level findings, and it attacks the right weakness: an individual horse is not measurable enough, while a jockey is. Twelve groupings were tested (§14).

- **Shrinking a horse toward its group beats shrinking it toward the population**, on every grouping tested — forward correlation 0.255 to **0.285** for the best, a 12% relative gain from changing nothing but the target. But *not* by the hypothesised mechanism: within every history-depth band the gain is at most ±0.02 and is *negative* for the thinnest horses. The gain is Simpson's paradox — the group term makes thin and thick horses comparable *across* depths, which is what a within-race model needs.
- **Volatility is a group property, and the group beats the individual.** Running style explains 20% of the between-horse variance in log volatility against a 0.07% noise floor, with a 1.55× spread between the steadiest and most erratic groups — and the group's volatility forecasts a horse's next deviation **better than the horse's own volatility history does** (+0.112 against +0.101). That is a strict improvement on the individual-level result, where volatility was real and worth nothing.
- **Pedigree is a decisive null.** Sire and damsire carry 0.0001 of the variance in relative performance — on 530 units each holding as many runs as a jockey, so this is a finding,

not a sample-size problem. Hong Kong imports are pre-screened and the handicap exists to equalise ability within a class; whatever pedigree contributes is already in the rating.

- **Cold start is real and priced.** Origin and import type predict a first-time starter ($z = 5.6$, $z = 5.3$) whom the production model cannot see at all. But a group's prior market error predicts a debutant's market error at r between -0.005 and $+0.004$. The market has it.

The practical answer

Twenty-one candidate blocks in total were built forward-only and tested against the production architecture on identical rows, folds and seeds, with a no-op control that returns **exactly zero** in every run.

One block cleared a development gate — group-shrunk ability on the prior-class group, at $t = -3.80$ with all five years improving and a mean of -0.00197, which is **44% of the production model's entire edge**. Taken to canonical scale (8 years, 5 seeds, 400 rounds) it **retained 43% of its effect and lost significance**: -0.00085 at $t = -0.90$, six of eight years improving, with two years degrading by more than the mean improvement.

So nothing is recommended for promotion, and the canonical step is what established that — the most plausible mechanism being capacity, since at 400 rounds the booster recovers much of what the feature supplies at 150. That is a general caution about every development-profile number in this repository and is more valuable than the feature would have been.

The pattern across all twenty-one blocks is the study's most useful single result: the rank correlation between how *novel* a candidate is and how much it *helped* is +0.39. The more novel the information, the less it helped. Marginal value at this effect size lies in sharpening what the model already uses, not in adding what it lacks.

And the pattern across the nine is the most useful single thing this study produced: the correlation between how *novel* a candidate is and how much it *helped* is +0.39, meaning **the more novel the information, the less it helped**. Marginal value at this effect size lies in sharpening what the model already uses, not in adding what it lacks. That is a coherent explanation for the reference project's own history: roughly four hundred candidate columns built, audited, and none promoted.

The finding that outlasts every candidate

Three candidates were taken from the development profile to canonical scale (§15). All three had been favourable; none survived.

Candidate	Development Δ	Canonical Δ	Retained
group-shrunk ability, prior class	-0.00197 ($t = -3.80$, 5/5 yr)	-0.00085 ($t = -0.90$)	43%
finishing band, recency-weighted	-0.00111	-0.00016	15%
finishing band, field-relative rank	-0.00122	$+0.00028$	−23%

Mean retention about **12%**, and the strongest development result — $t = -3.80$ with every single year improving — retained the most while still losing significance. Two mechanisms point the same way: at 400 rounds the booster extracts from the existing 28 columns what the feature supplied at 150, and fifteen development cells at this effect size admit a t of -1.3 from sampling alone.

At this effect size the development profile has close to no predictive value for the canonical outcome. That matters because the development profile is exactly the gate the reference project uses to decide which candidates earn a canonical run. Its workflow is not wrong — it requires canonical confirmation before promotion — but the screening stage should be understood as a filter on gross defects and schema correctness, not as evidence about a candidate's value.

The concrete implication: stop generating candidates and screening them cheaply. The only measurement that carries information here is the expensive one, so the right strategy is far fewer candidates, each taken straight to canonical scale.

Where this leaves the three original recommendations

All three were tested at the production level in §14, and two are now closed negatively despite being real as measurements.

1. **Recency-weighted ability at the measured 60-day half-life:** +0.00007. Nothing. The forward-correlation profile that motivated it is unambiguous, and the production model evidently already recovers it.
2. **The within-card track-speed signal:** +0.00041 — worse — and −0.000005 in its style-interacted form. It correlates 0.50 with a race's own shock and still cannot move a win probability, which is what the mechanism predicts: a shock common to the field cancels from the ordering, and the differential loading through style is too small to matter. Worth knowing before anyone spends a leakage-rule exception on it.
3. **The sectional finishing block:** −0.00087 at $t = -1.07$, unchanged and still the second-best block measured. It remains the one candidate whose canonical confirmation has not been spent, and is the single most defensible next run.

2 The existing project and the data

2.1 What the reference system does

The reference project estimates a calibrated probability that each declared runner wins, for HKJC Class 1–5 races at Sha Tin and Happy Valley. Its production estimator does not learn a probability; it learns a *correction* to the market's own. For race r and runner i, with q_{ri} the market's normalised implied probability,

$$u_{ri} = \log q_{ri} + c_\theta(x_{ri}), \qquad p_{ri} = \mathrm{softmax}_i(u_r),$$

where $\log q$ enters XGBoost as a fixed additive base margin and the booster learns c_θ against the racewise winner-softmax likelihood. At $c = 0$ the model *is* the market exactly, because q already sums to one within the race. Every published improvement is therefore an improvement over the market on the same races.

Its measured standing, on 7,933 races over 2016–2026: the production five-seed ensemble beats the market by -0.004533 of race log loss in ten of eleven years. The unanchored baseline beats it by -0.000992 but changes sign in four years. That contrast — anchoring converts a coin flip into a consistent one — is the architecture's argument.

One prior result is directly relevant here and is the strongest existing evidence for the framing this study adopts. Replacing independent Gumbel choice shocks with independent normal ones, holding the mean utilities fixed, improves race log loss by -0.0126 standalone and -0.00094 market-relative, in eleven of eleven years. **Latent performance with normal noise is already known to describe this sport better than the softmax the production model uses.**

2.2 The archive, and the measurement that makes this study possible

Fourteen scraped datasets are available. This study uses five: `sectional-times` (192 MB) for per-section times, `aspx-results` for prices, weights, draw and human connections, the `horses` per-start JSON for the official rating and declared gear (which the result pages do not carry), `horse-profiles`, and `comments-on-running` for the stewards' text.

Table 1: Sectional geometry, measured on every season. Sections after the first are exactly 400 m, so indexing them *from the finish* makes them physically comparable across every distance. Valid on 1.0000 of races.

Distance (m)	Sections	Section lengths, start to finish
1000	3	200, 400, 400
1200	3	400, 400, 400
1400	4	200, 400, 400, 400
1600	4	400, 400, 400, 400
1650	4	450, 400, 400, 400
1800	5	200, 400, 400, 400, 400
2000	5	400, 400, 400, 400, 400
2200	6	200, 400, 400, 400, 400, 400

Coverage is close to complete: over 648,121 section rows, `sectional_time_sec` is missing on 0.0000 in every one of nineteen seasons, finishing time on 0.0002–0.0017, and the per-section running position on 0.0000–0.0004. Each row also carries the leader's own time for that section, the margin behind the leader, and — for sections after the first — 200 m sub-splits, which this study has not exploited.

2.3 Population, and reconciliation against the reference

A race enters if it is Class 1–5 at one of the two Hong Kong tracks, every runner is priced, the reciprocal-odds sum is at least 1.10, the field has at least four runners and the finishing time is physically plausible. Races are kept whole or dropped whole, because a partial field breaks every within-race normalisation.

Result: **165,435 runner-races over 13,309 races**, 2 April 2008 to 15 March 2026, mean overround 1.2291.

Restricting to 2010 onward to match the reference training start, 12,041 of its 12,237 races are shared, and on 149,494 jointly identified runner rows the market probability computed here agrees with the reference table's to 5.55×10^{-17} at the 99th percentile — the same number to float64 noise.

The residual disagreement is understood rather than tolerated. Exactly 72 races (0.60%) have one fewer runner here, never more, and the missing runners are `UR`, `PU` and `WV` codes: **the sectional archive records no times for a non-finisher**. The reference project deliberately keeps non-finishers in the pre-race field as losing starters, because removing them by post-race status is field-definition look-ahead. This study's performance panel is therefore conditioned on finishing, which is a selection effect carried into §17.

3 Constructing a performance target

3.1 Model 0: what the environment explains

A ridge regression of performance on one-hot environment blocks, with no horse, jockey or market column in the design — a par that knew who was running would absorb the ability it exists to measure. Two versions are kept apart: an in-sample fit used only to decompose variance, and a forward fit, refitted for each evaluation year on strictly earlier years, used for every predictive claim.

Table 2: Nested in-sample R^2. Almost all of raw *time* is distance; the interesting variation lives in speed.

Block	R^2 (seconds)	ΔR^2	R^2 (log speed)	ΔR^2
distance	0.994597	0.994597	0.661373	0.661373
+ venue × surface	0.994831	0.000234	0.677483	0.016110
+ going	0.995291	0.000460	0.707558	0.030075
+ course & rail	0.995312	0.000020	0.708825	0.001267
+ class	0.995781	0.000470	0.740827	0.032002
+ field size	0.995795	0.000014	0.741852	0.001025
+ season	**0.996177**	0.000382	**0.767558**	0.025706

Forward par R^2 on log speed averages 0.7479 across evaluation years, so the par model generalises about as well as it fits.

3.2 Beaten lengths are not a measurement

The plan lists "beaten-length-adjusted performance" as a candidate target. One measurement removes it. Regressing beaten seconds on beaten lengths through the origin gives a slope of **0.15996 s per length**, and the same value to four decimals on every distance from 1000 m to 2200 m — that is exactly 6.25 lengths per second. Testing the stronger claim:

$$\text{lengths} = \text{round}\big(\text{beaten seconds} \times 6.25 \times 4\big)/4$$

holds exactly on 83.7% of 148,575 rows and to within a quarter length on 99.7%; the median ratio is 6.250000.

`LBW` is therefore the recorded time restated in quarter-length units, and carries no information beyond it. One consequence reaches the reference project directly: its `margin_rank` state family — eight features built from "99.8%-covered parsed `LBW` histories" — and its `dynamic_speed` family are measuring the same physical quantity, and any apparent complementarity between them is quantisation noise.

3.3 Q1: which target

Targets are compared on *forward persistence*: does a horse's date-atomic prior mean predict its next realisation. A target a horse cannot repeat carries no ability signal to find.

Within-race normalisation wins, and the reason is structural: it removes the race-level shock, which is 32% of the absolute residual's variance and is pure noise for the purpose of estimating a horse's ability. Both targets are carried forward, because the absolute one is the only one that can answer a question about absolute environmental effects.

Table 3: Forward persistence of each candidate target, $n = 104{,}539$.

Target	Forward r	Forward R^2	Split-half reliability
y_z^{rel} within-race z of log speed	**0.2777**	**0.0771**	0.822
y^{rank} within-race rank percentile	0.2643	0.0698	0.840
y^{par} par residual, log speed	0.2376	0.0565	0.557
y_z^{par} standardised par residual	0.2357	0.0556	0.579
y^{par} par residual, seconds	0.2195	0.0482	0.460
y^{beaten} beaten lengths	0.2132	0.0454	0.431

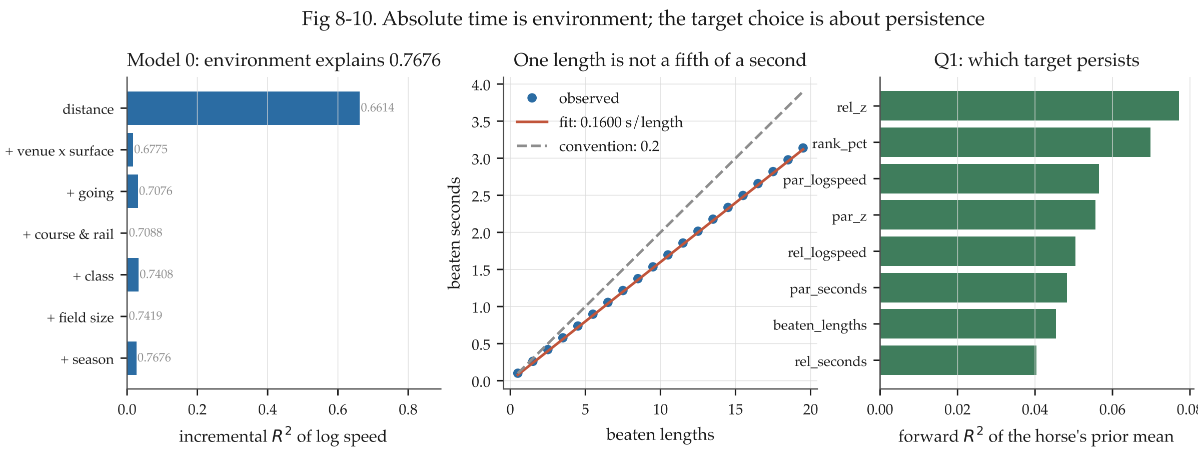

Figure 1: Model 0 and the target. Left: incremental R^2 of log speed by environment block. Centre: beaten seconds against beaten lengths — the fit is 0.15996 s per length, not the conventional 0.2, and the relationship is arithmetic rather than statistical. Right: forward persistence of each candidate target.

4 Exploratory analysis: what is identifiable

Table 4: History depth per estimation unit. This table determines what the rest of the study can and cannot answer.

Unit	Units	Median	< 5	< 10	≥ 20	Max
Starts per horse	8,599	15	16.3%	33.0%	41.0%	97
Rides per jockey	190	121	19.5%	27.9%	62.6%	9,980
Rides per horse–jockey pair	68,454	**1**	**87.8%**	97.5%	0.23%	45

The third row is decisive. A horse–jockey interaction estimated from a median of one observation per cell is not an estimate, and §10 confirms it formally rather than asserting it.

Horse × environment cells are thin but not hopeless: the median horse–going cell holds 3 starts and 59.6% hold fewer than five, while the median horse–venue cell holds 8 and 42.2% reach ten or more. That difference is why §7 finds a persistent venue sensitivity and no persistent going sensitivity, and it is a difference in *measurability* as much as in nature.

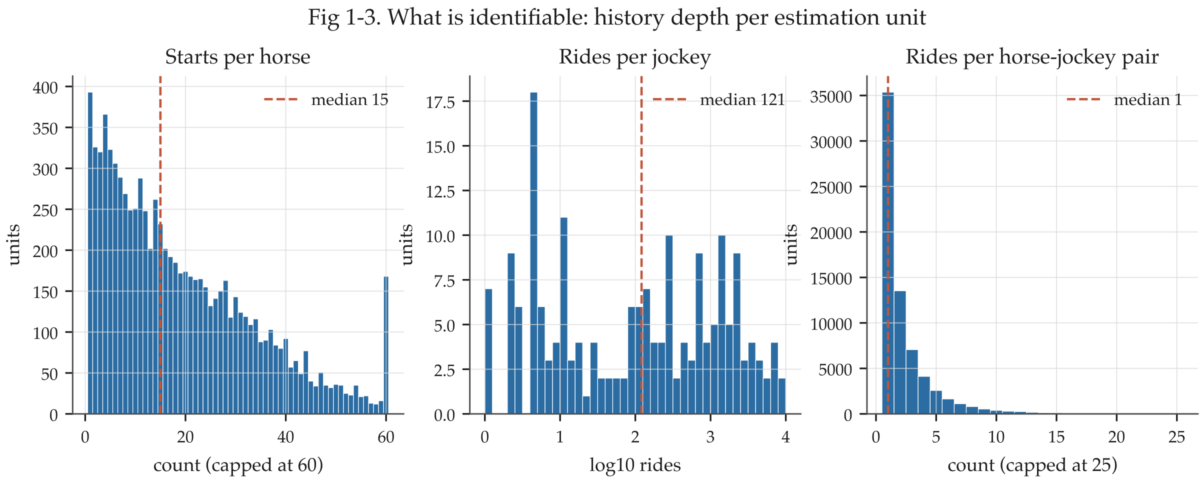

Figure 2: History depth per estimation unit. The right-hand panel is the constraint on Q4.

Fig 4. Why partial pooling is not optional

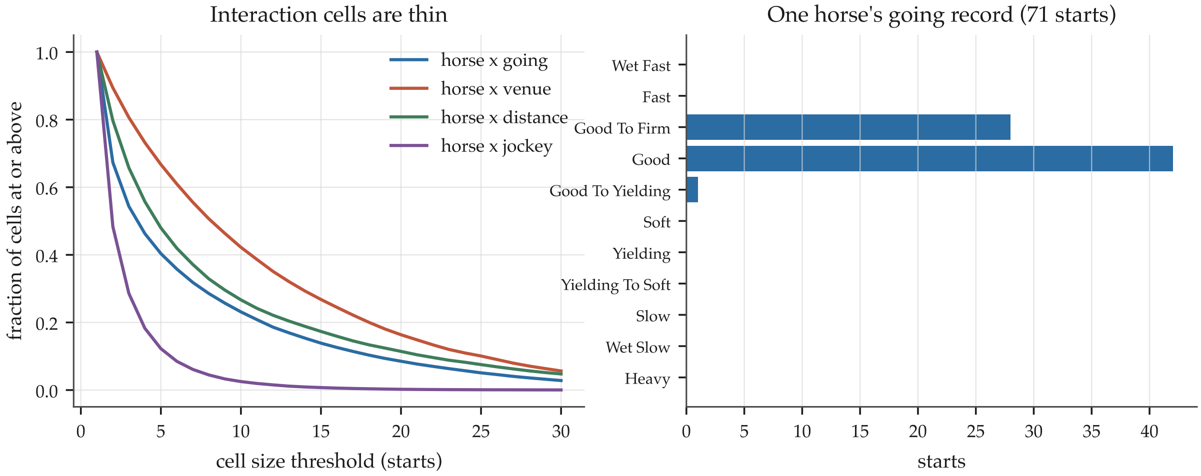

Figure 3: Why partial pooling is not optional. Left: the fraction of interaction cells reaching each size. Right: one well-raced horse's record across going levels.

5 Horse and jockey latent effects

5.1 Estimation

Crossed random effects are estimated by EM on the mixed-model equations: the E step is the best linear unbiased predictor at the current variance ratios, solved by Gauss–Seidel backfitting with `np.bincount`, and the M step is the variance-component update that adds each effect's posterior variance to its squared mean. Omitting that posterior-variance term is the classic error and biases every τ^2 downward.

The estimator is verified rather than trusted. On simulated crossed data with $\tau_{\text{horse}} = 0.55$, $\tau_{\text{jockey}} = 0.20$, $\sigma = 0.85$ it recovers 0.544, 0.192, 0.851. On data where the jockey factor has *no* effect it returns $\tau_{\text{jockey}} = 0.0093$, a variance share of 8.5×10^{-5} — so an effect it reports is not an artifact of the machinery.

5.2 Q2 and Q3: variance components

Table 5: Variance shares on within-race relative performance.

Model	Horse	Jockey	Trainer	Residual
1: horse	23.3%	—	—	76.7%
2: + jockey	18.4%	3.7%	—	78.0%
2b: + trainer	17.7%	3.5%	1.2%	77.7%
3: + observable covariates	17.6%	3.0%	1.1%	78.3%
Jockey alone, no horse control	—	6.0%	—	94.0%
Trainer alone	—	—	1.4%	98.6%

Controlling for the horse removes about **39%** of the apparent jockey variance (6.0% to 3.7%). That is the size of the assignment channel: good jockeys ride good horses.

5.3 The surprise: jockey identity forecasts better than horse identity

Table 6: Forward evaluation on the confirmation window, 2020–2026. The scale is fitted on the previous year's own earlier fit, so it never sees the scored year.

Model	oos R^2 (scaled)	Forward scale	Mean r	Years positive
1: horse	0.0124	0.39	0.141	7/7
Jockey alone	**0.0479**	**1.05**	0.223	7/7
2: horse + jockey	0.0496	0.66	0.227	7/7
2b: + trainer	0.0536	0.67	0.236	7/7
3: + covariates	**0.0582**	0.64	0.245	7/7
Trainer alone	0.0063	0.85	0.094	7/7

Jockey identity alone forecasts nearly four times as well as horse identity, and at a forward scale of 1.05 — essentially perfectly calibrated — where the horse effect needs shrinking to 0.39 of its fitted size. Adding the horse to the jockey improves oos R^2 from 0.0479 only to 0.0496.

The explanation is precision and drift, not size. A jockey effect rests on roughly a thousand rides and does not move; a horse effect rests on fifteen starts and does. Two consequences worth separating:

- As a *predictor*, jockey identity is powerful partly *because* it encodes assignment — the market knows this too, which is why the production model's `jockey_residual` is a residual *against the market* rather than a strike rate.
- As a *causal* effect it is much smaller. The horse-adjusted share is 3.7%, and the plan's own instruction not to read a jockey random effect causally without discussing assignment bias is exactly right.

5.4 Identification diagnostics

Because jockey assignment is highly non-random, the effect is estimated a second, independent way: from *within-horse* variation only, using each ride's deviation from that horse's own mean.

The within-horse estimates have standard deviation 0.141 against a median standard error of 0.028 — **5.05 times their own noise**, so they are not sampling error — and they agree with the crossed BLUPs at Spearman **0.970**. Horse control shrinks the spread of jockey effects by 17.7%, and nine of the top ten jockeys by raw average remain in the top ten after adjustment.

5.5 Q2: ability drifts, with a two-month half-life

The ordering is monotone with no interior structure to over-read: recent form matters far more than distant form, and the whole career is the worst estimator tested. This speaks directly to the reference project's open backlog item on recency-weighted training, where truncating history was found to hurt badly overall while helping recent years — weighting rather than truncating is the untried alternative, and the optimum here is a 60-day half-life.

Table 7: Forward correlation of a decayed horse-ability estimate. Correlation is the criterion because it is scale-free; the optimal scale column shows how badly a static estimate is over-scaled.

Recency half-life	Forward r	R^2 at best scale	Implied optimal scale
30 days	0.1445	0.00065	0.870
45 days	0.1485	0.00244	0.772
60 days	**0.1497**	**0.00314**	0.716
90 days	0.1488	0.00325	0.650
180 days	0.1407	0.00106	0.550
365 days	0.1301	−0.00201	0.470
730 days	0.1227	−0.00406	0.422
Static, all history	**0.1142**	−0.00628	**0.369**

Fig 18-20. Horse and jockey latent effects

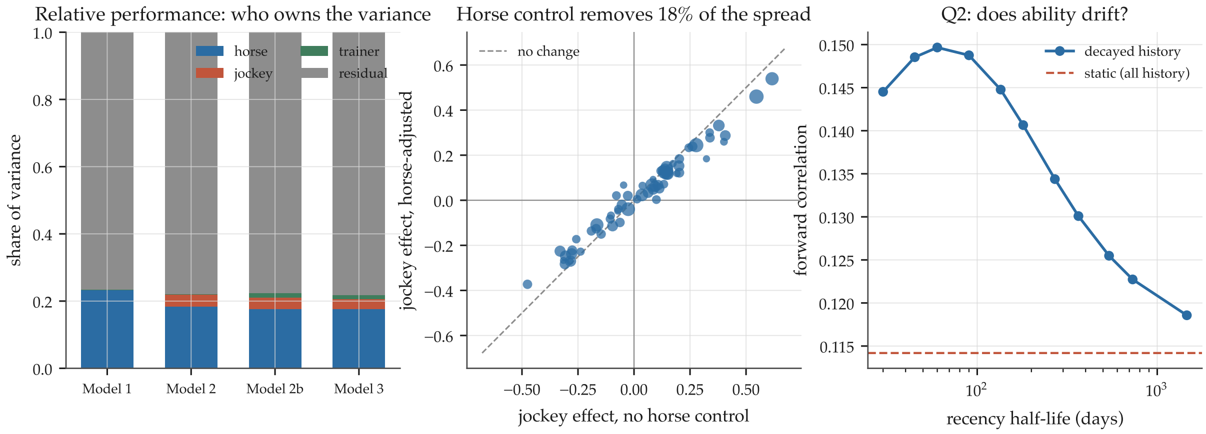

Figure 4: Left: variance shares by model. Centre: the jockey effect before and after horse control, point size proportional to rides. Right: forward correlation against recency half-life; the static estimate is the worst of the eight tested.

Fig 21-22. Horse ability and its uncertainty

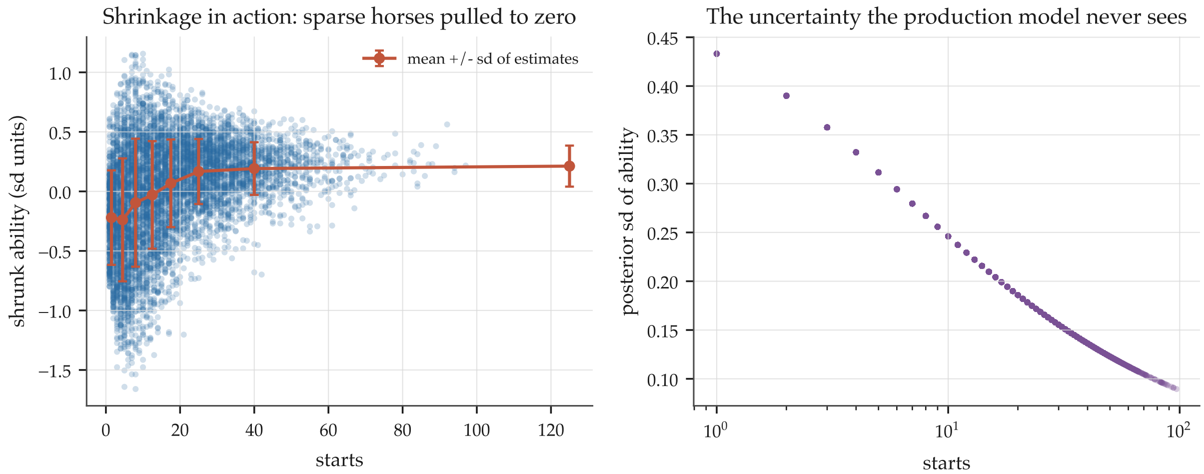

Figure 5: Shrunk horse ability against starts, and the posterior standard deviation of the estimate. The right-hand panel is a quantity both of the reference project's latent engines compute and neither promotes.

6 Environmental effects: absolute time against relative ordering

The plan's Q5 asks for both quantities separately, because they are routinely confused and they point in opposite directions here.

On absolute time, the environment dominates. It explains 0.9962 of the variance of raw seconds. Holding distance, venue and surface fixed at Sha Tin turf 1200 m, going alone moves mean finishing time by **1.196 s** across adequately-supported levels — roughly seven and a half lengths.

On relative ordering, almost nothing survives. After the forward par model, the residual race-level shock still has a standard deviation of 0.00724 in log speed (about half a second at 1200 m), and the observable card explains almost none of it:

Table 8: One-way ANOVA R^2 of the residual race-level shock, count-weighted.

Dimension	Levels ($\geq$ 200 races)	ANOVA R^2	Range in seconds at 1200 m
going	3	0.0298	0.221
venue	2	0.0099	0.102
distance band	4	0.0058	0.116
class	4	0.0008	0.050
course letter	4	0.0005	0.035
surface	2	0.0004	0.032

So roughly **96% of the race-level shock is unobserved** from the card — day-to-day track speed, weather and tactics. §11 shows that most of it *is* observable, just not from the card: it is observable from the rest of the meeting.

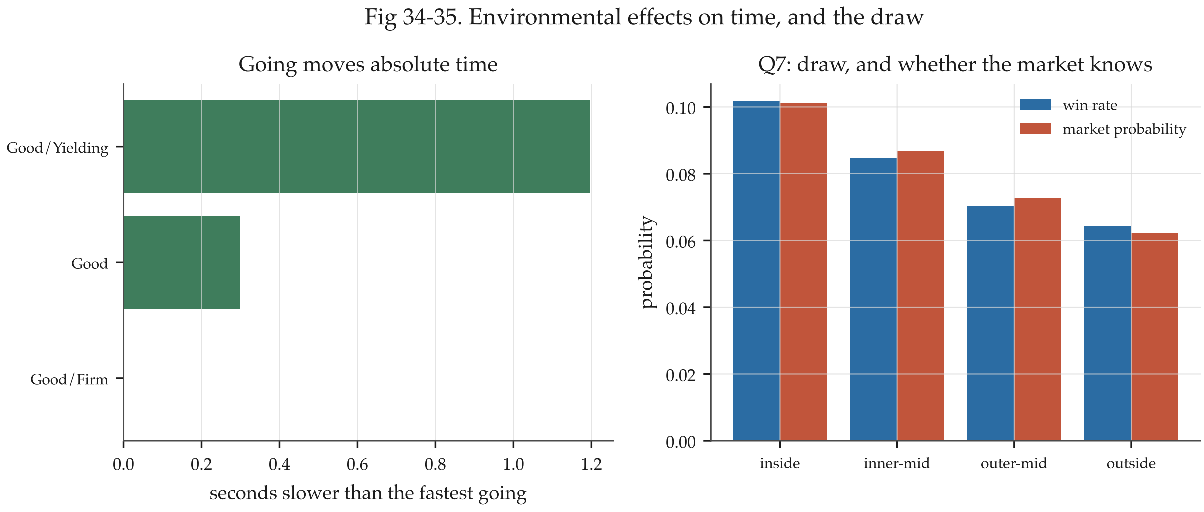

Figure 6: Left: going against absolute finishing time on one distance, track and surface. Right: draw against the win rate and against the market's price for the same runners.

7 Horse × environment effects

The plan calls this the single most important question, and prescribes the right test: what matters is not a horse's apparent sensitivity but the between-horse variance of the *true* sensitivity, and the only evidence for that is whether an early estimate predicts a later realisation.

The design is deliberately hard to fool. For each binary environmental contrast, form each horse's performance contrast from its *early* starts; decompose the spread of those estimates into true between-horse variance and sampling noise by method of moments; then test the early estimate against the same horse's *later* contrast.

Table 9: Horse × environment sensitivity. "Real share" is the fraction of the apparent spread that is not sampling noise. Persistence is the correlation between the early and later estimates, with z against zero.

Contrast	Horses	Apparent sd	Real τ	Real share	Persistence
going, soft vs firm	1,675	0.515	0.216	0.175	+0.034 ($z = 1.4$)
venue, HV vs ST	1,476	0.599	0.374	0.389	+0.215 ($z = 8.3$)
surface, AWT vs turf	365	0.607	0.402	0.438	+0.338 ($z = 6.4$)
distance, long vs short	163	0.634	0.387	0.374	+0.064 ($z = 0.8$)
field size, large vs small	36	0.608	0.306	0.254	−0.138 ($z = -0.8$)
class, high vs low	—	—	—	—	unanswerable

Three readings, in order of importance.

Most apparent sensitivity is noise, everywhere. Even for the contrasts that persist, the real between-horse standard deviation is 0.37–0.40 against an apparent 0.60 — so 56–62% of what a naive per-horse table would show is sampling error. A study that stopped at the apparent spread would have reported five real effects.

Going sensitivity does not persist. This is the plan's own nominated "most valuable possible finding" and it is confirmed: going moves absolute race time by 1.2 s, and horse-specific going sensitivity has a forward correlation of 0.034 at $z = 1.4$ on 1,675 horses — the largest sample of the five contrasts. The effect on *everyone* is enormous; the differential effect is not measurable, and is not usefully there.

Venue and surface sensitivity are real. "Horses for courses" between Happy Valley's tight circuit and Sha Tin's galloping track survives at $z = 8.3$ with a true between-horse standard deviation of 0.374 performance standard deviations — a materially large effect. Surface sensitivity is stronger still ($z = 6.4$, $\tau = 0.402$) on a much smaller qualifying sample.

The measurability difference tracks the cell counts exactly: the median horse–venue cell has 8 starts against the median horse–going cell's 3. Part of this result is about nature and part is about what 13,309 races can resolve, and the honest statement separates them.

Fig 31-33. Horse x environment: apparent, real, and persistent

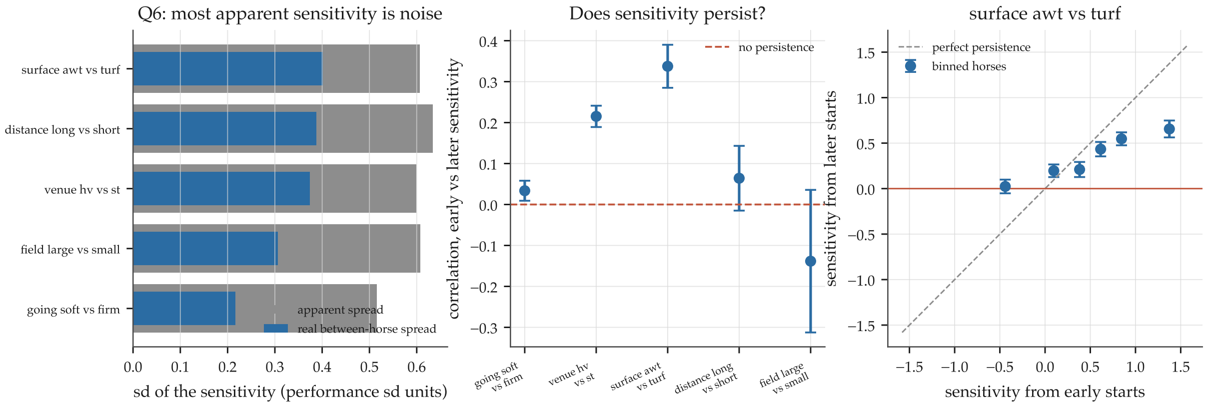

Figure 7: Left: apparent against real between-horse spread for each contrast — the gap is sampling noise. Centre: persistence with its null standard error. Right: early against later sensitivity for the strongest contrast, binned.

8 The residual distribution

Measured on 96,579 genuinely out-of-sample residuals from a forward-fitted mean model with the forward scale applied, 2016–2026.

Table 10: Shape of the standardised out-of-sample residual.

Statistic	Value	Normal
Skewness	−0.729	0
Excess kurtosis	+0.810	0
IQR / sd	1.1195	1.3490
Mean − median	−0.070	0

The centre is *more* concentrated than a normal and the tails are heavier — IQR/sd of 1.12 against 1.35 — with a pronounced left skew. Candidate densities were fitted on the development window and scored on the confirmation window by predictive log-likelihood, so no family is preferred for fitting its own data.

Table 11: Held-out predictive log-likelihood per observation, relative to the Normal.

Family	Parameters	Gain over Normal per observation
Two-component normal mixture	5	+0.0557
Skew-t	4	+0.0553
Skew-normal	3	+0.0455
Student-t ($\nu = 5.18$)	3	+0.0246
Normal	2	0

8.1 Which tail, and what it was

The plan asks whether the bad-performance tail is materially heavier, and whether the outliers are identifiable events. Both answers are yes, and the second is checkable because the reference archive holds the stewards' text.

Table 12: Stewards'-comment rates by residual band. "Severe" matches the language used for a run that was stopped or abandoned; its base rate is 9.5%.

Residual band	n	Any-trouble rate	Severe rate	Severe lift
< -3 sd	889	0.693	**0.460**	**4.83**
-3 to -2	4,200	0.624	0.259	2.71
-2 to -1	10,367	0.608	0.157	1.65
-1 to $+1$	70,122	0.575	0.078	0.82
$+1$ to $+2$	10,240	0.501	0.057	0.59
$+2$ to $+3$	755	0.437	0.050	0.53

The gradient is monotone across all bands in both tiers. Nearly half of the runs three standard deviations below expectation carry a severe stewards' phrase, at 4.8 times the base rate — and good performances are *less* likely than average to carry one.

Two interpretive points the plan explicitly asks for. First, these observations are *not* removed: a disastrous run is part of the outcome distribution a probability model must represent, and the reference project's own settlement treats non-finishers as losing starters for the same reason. Second, part of the left skew is **structural rather than stochastic**: a beaten horse is eased, so the lower tail of a within-race time distribution is long by the physics of the sport. The mixture's fitted "bad" component carries probability 0.44, which is far too large to read as a bad-trip rate and is better read as the asymmetry of the racing outcome itself.

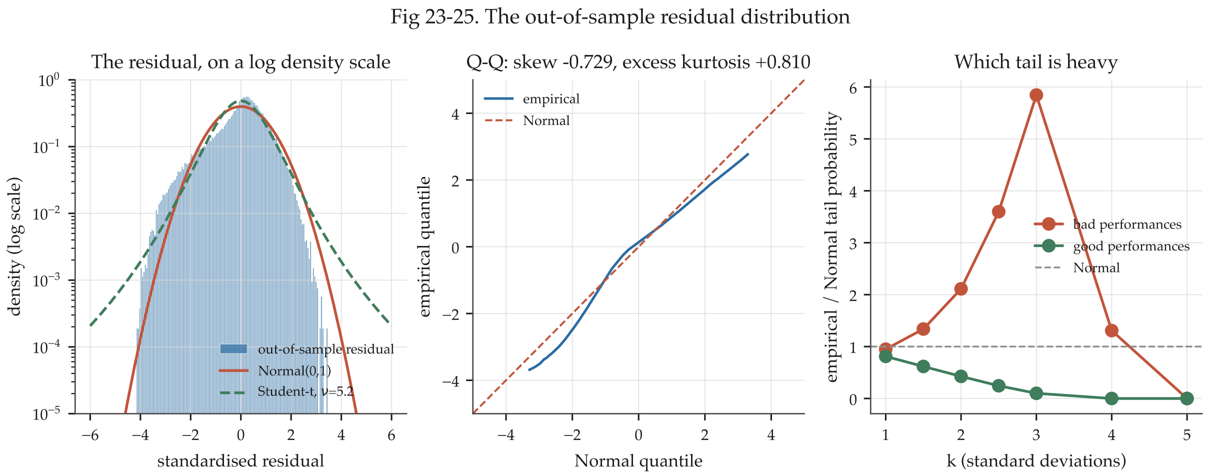

Figure 8: Left: the residual on a log density scale with Normal and Student-t overlays. Centre: the Q–Q plot. Right: empirical over Normal tail probabilities, split by sign — the bad tail is the heavy one.

Fig 26-27. Choosing a density, and what the outliers were

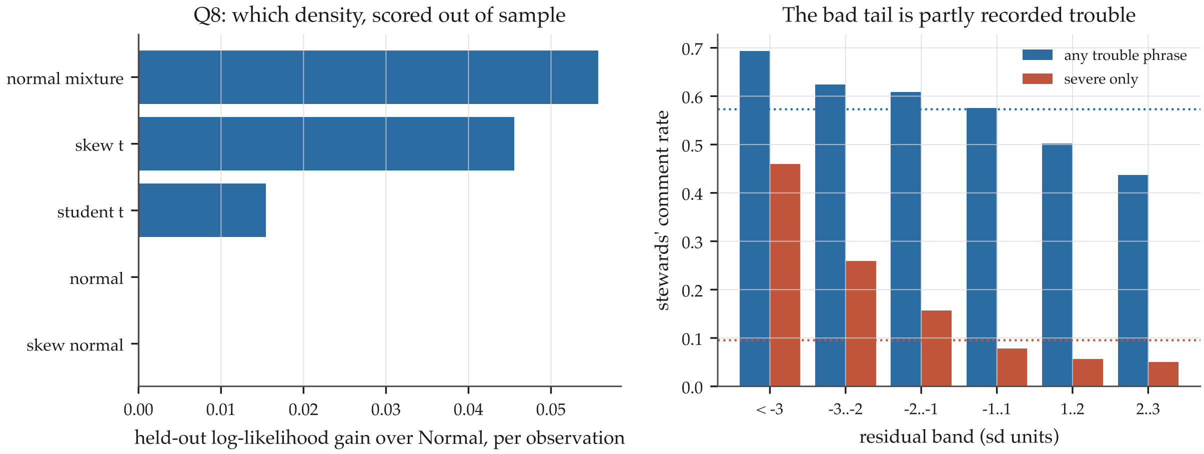

Figure 9: Left: held-out log-likelihood gain over the Normal. Right: stewards'-comment rates by residual band, two severity tiers.

9 Heteroskedasticity and horse-specific volatility

The plan judges constant variance more important than exact normality, and it is right: a model with the mean right and the spread wrong is miscalibrated where race log loss punishes hardest.

9.1 Q9: the variance is not constant

Table 13: Residual spread by decile of each covariate.

Covariate	sd, lowest decile	sd, highest decile	Ratio
Market rank	0.812	1.051	**1.295**
Market probability	1.051	0.824	0.784
Draw position	0.914	0.984	1.077
Field size	0.933	0.949	1.017
Carried weight	0.960	0.936	0.975

The largest driver is the market's own assessment, and the direction is informative: **favourites run more consistently**. A joint model of $\log|e|$ on observables reaches out-of-sample R^2 0.011 with an implied p90/p10 spread ratio of 1.385.

9.2 Q10: volatility is a horse trait, and it persists

The naive per-horse standard deviation is dominated by sample size, exactly as the plan requires be demonstrated. Decomposing the observed variance of per-horse log sd:

The decisive test is the plan's own: split each horse's residual history at its own median date and ask whether the first half's spread predicts the second's.

The correlation rises with depth, which is what a real trait measured with less noise should do. As a forward feature, a prior-only volatility estimate predicts the next start's $|residual|$ at Spearman $+0.087$ to $+0.122$, with the realised spread ratio between the top and bottom terciles reaching 1.255 at twelve prior starts.

Table 14: Variance components of per-horse log residual standard deviation, $n = 5{,}324$ horses with at least two residuals.

Quantity	Value
Observed variance of $\log \hat{\sigma}_i$	0.2060
Mean sampling variance $\approx 1/(2(n_i - 1))$	0.0786
True between-horse variance τ^2	0.1274
Share of the observed spread that is real	**61.8%**
τ (sd of true $\log \sigma$)	0.357
Spread ratio for a horse 1 sd more volatile	**1.43**×

Table 15: Split-half persistence of per-horse log residual sd.

Min starts each half	Horses	r	z vs zero	Later-sd ratio, top/bottom tercile
4	3,876	+0.144	+8.94	1.115
6	3,146	+0.156	+8.77	1.119
8	2,519	+0.176	+8.85	1.131
10	2,015	+0.170	+7.63	1.115
15	1,024	+0.223	+7.14	1.148

So horse-specific volatility passes every test the plan sets for it. Whether it is *useful* is a separate question, and §12 and §13 answer it differently — which is the most instructive disagreement in this study.

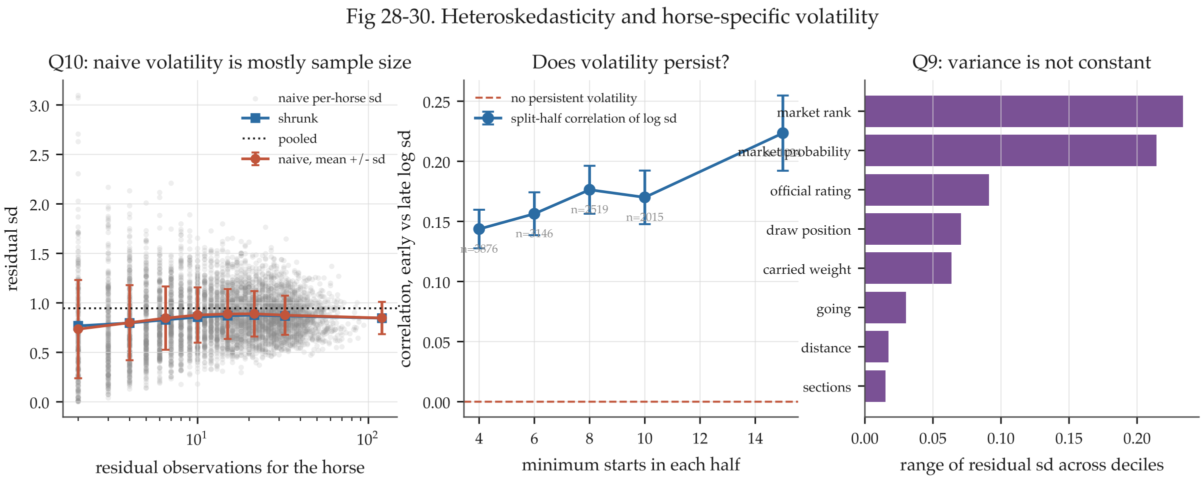

Figure 10: Left: naive against shrunk per-horse volatility by observation count, with the pooled value. Centre: split-half persistence with its null standard error. Right: the covariates that move the spread.

10 Horse × jockey compatibility

10.1 A correction, recorded

The first version of this analysis reported early-to-late pair correlations of −0.04 to −0.21 with z from −3.8 to −7.4 — an apparently strong *negative* result that would have read as compatibility mean-reverting.

It was an artifact of the residualisation, and the mechanism is worth stating because it is easy to reproduce. The horse and jockey main effects were fitted over the whole sample, so a horse's effect is roughly the average of its early and late performance. The early pair residual then becomes about (early − late)/2 and the late one about (late − early)/2: forced anticorrelation with no compatibility involved. The artifact was *larger in magnitude* than a true pair standard deviation of 0.30 would have produced, which is how it was caught.

The corrected design fits the main effects separately within each era, so the early estimate and the late realisation share no fitted parameter.

10.2 The corrected answer, with power

Table 16: Forecastability of a pair effect, with the correlation a real effect of each size would have produced on the same design.

Min rides each half	Pairs	Apparent sd	True τ	r	z
2	575	0.483	0.234	−0.002	−0.05
3	249	0.383	0.155	+0.024	+0.38
4	121	0.303	**0.000**	+0.071	+0.77
5	61	0.219	**0.000**	−0.053	−0.41
Power: what a real pair effect would have shown					
true pair sd = 0.10				+0.058	
true pair sd = 0.20				+0.211	
true pair sd = 0.30				+0.334	

Meanwhile the **in-sample** Model 7 fit, on pairs with three or more rides, reports $\tau_{HJ} = 0.298$ and a **10.3%** variance share.

That contrast is the finding. An in-sample pair effect apparently worth a tenth of the variance, and a forecastable component indistinguishable from zero on a design with the power to see half that size. Q4 is answered negatively, and the negative is supported rather than merely reported.

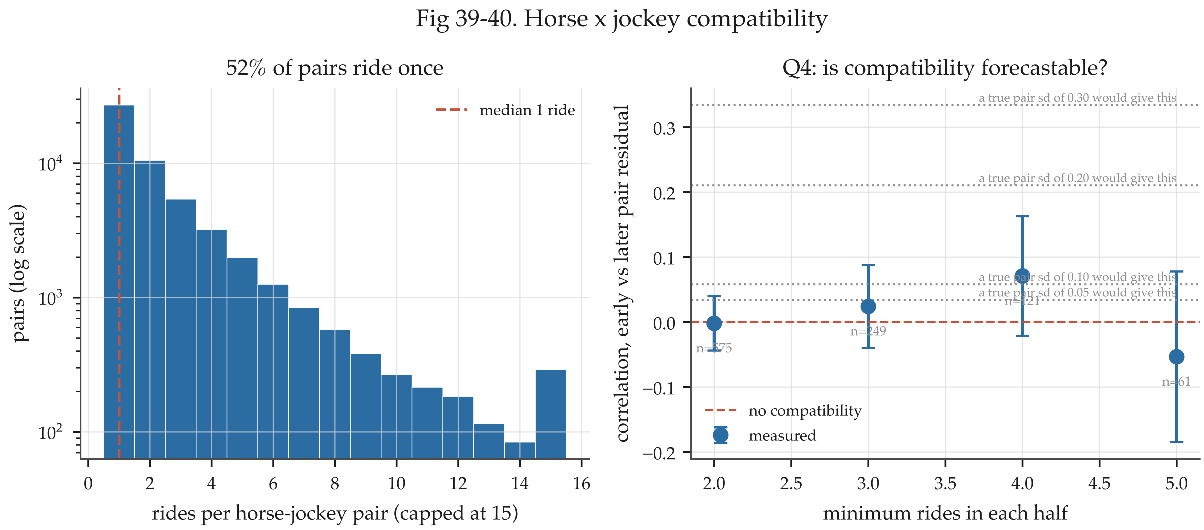

Figure 11: Left: rides per horse–jockey pair on a log count scale — 52% ride once. Right: measured persistence against what a real effect of each size would have produced.

11 Shared race effects and pace

The reference project already tried a one-factor structured-covariance probit and got a negative result, with the factor scale essentially unidentified: its grid selected “no factor” in seven of eleven years and the whole range from 0.00 to 0.80 spanned about 2×10^{-5} of log loss. Two features of that test left the question open. Its loading was a running-*position* habit parsed from the result page’s three-number line rather than a timed pace measurement, and its standardisation used the whole frame including the test block.

11.1 The shock is large, and mostly a property of the day

The shared shock is **32.1%** of the absolute residual’s variance, with a standard deviation of 0.00724 in log speed — about **0.503 s at 1200 m**.

Table 17: How predictable a race’s own shock is.

Predictor	Correlation with the race’s shock
Running mean of earlier races on the same card	**0.504**
by the 8th race of the meeting	**0.584**
Prior history of the same venue and surface	0.179

Track speed is therefore a day-level property and it is largely *measurable* before a later race on the card runs. This is the one signal in this study that is both large and structurally unavailable to the reference pipeline, whose point-in-time rule excludes all same-date information. That rule exists to stop a horse seeing its own same-day outcome; a track-speed reading taken from race one before race eight goes to post is forward in time and is not that. It is recorded here as a labelled deviation and is not folded into any candidate feature.

11.2 Q11: the shock does load differently by running style

A shock loading equally on every runner cancels from every pairwise utility difference and cannot change any probability, so the interaction is the whole question. Measuring the race's pace directly from the sectional archive:

Table 18: Pace-by-style interactions on within-race relative performance, development window fitted, confirmation window scored.

Specification	Interaction coefficient	z	oos R^2 gain
Forwardness × early pace	−0.00034	−8.31	−0.00002
Forwardness × pace shape	+0.00036	+8.55	+0.00035
Closing ability × early pace	+0.00031	+7.43	+0.00118

The signs are the physics: a fast early pace *hurts* habitual front-runners and *helps* closers. So the mechanism the reference project's one-factor probit hypothesised is real and correctly signed — it was undetectable with a loading that could not see pace. Its economic size is small.

11.3 Is the pace forecastable, and does the mispricing survive?

Everything above conditions on the pace the race actually ran. A model can only use a pace it can project.

Table 19: Out-of-sample forecastability of a race's early pace.

Projection	oos r	oos R^2
Top-3 prior style only	0.032	−0.005
Field style composition (12 features)	**0.414**	**0.158**
+ earlier races on the same card	0.405	0.153
Earlier races on the same card only	0.098	−0.003

A naive summary sees almost nothing; a proper projection recovers $r = 0.41$. Note also that track speed and early pace are largely *distinct* quantities — card track speed alone predicts early pace at only 0.098 — so the surface's speed and how hard they went early are separate facts.

The market's error across the style × pace grid swings −0.0398 on the *realised* pace, and −0.0151 on the *projected* pace: **37.8% survives**, with a maximum absolute residual of 0.0087 on 50,784 confirmation-window runners. Roughly nine tenths of a percentage point of win probability, available pre-race, at the extremes of the grid.

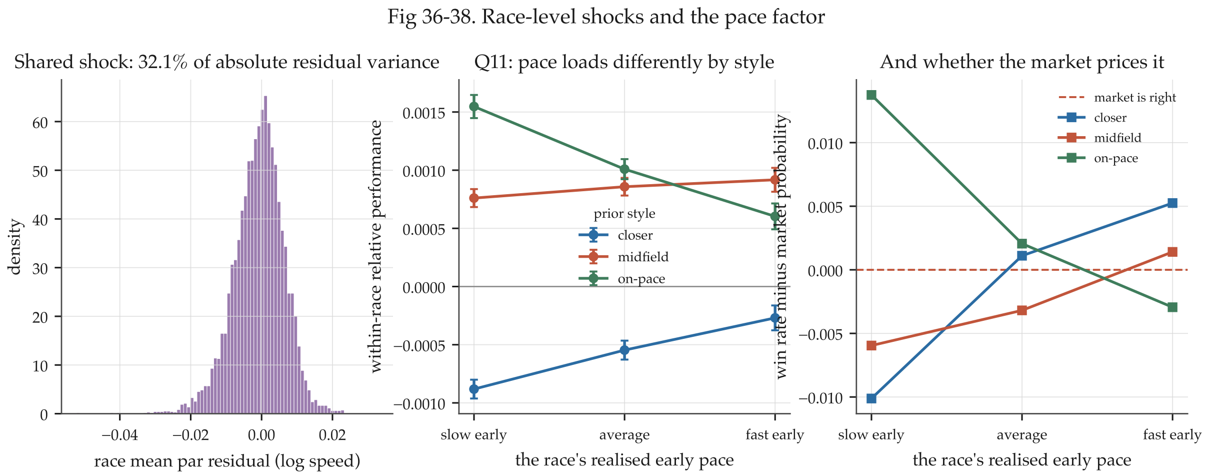

Figure 12: Left: the distribution of the race-level shock. Centre: relative performance by prior style and realised early pace — the interaction is visible. Right: the market residual over the same grid.

12 From latent performance to probabilities

12.1 Integration, and why not simulation

For independent shocks the winner probability collapses to a one-dimensional integral over the winner's own shock,

$$P(i \text{ wins}) = \mathbb{E}_z \left[\prod_{j \neq i} F_j \left(\frac{\mu_i - \mu_j + \sigma_i z}{\sigma_j} \right) \right],$$

which holds for *heterogeneous* σ as written. Simulation is avoided for a reason the reference project measured: a winner-counting estimator over m draws cannot resolve a probability below $1/m$, and this sport's longshots sit at 10^{-3} to 10^{-4} where race log loss is decided.

The integrator is verified against every case with a known answer: two runners at equal scale to 1.1×10^{-16}, at unequal scale to 1.3×10^{-12}, an identical twelve-runner field to 1.4×10^{-17} from uniform, a longshot resolved at 2.3×10^{-5}, total mass to 1.1×10^{-16}, a constant shared-factor loading collapsing onto the iid answer exactly, and the Student-t path against a 400,000-draw simulation to within its simulation standard error.

12.2 Q13: what each piece of structure buys

Six nested variants, each a controlled comparison against the row above. One global utility scale is fitted per variant on the last development year and applied forward; that year is scored on nothing.

No elaboration beats plain homoskedastic normal shocks. This is the central negative result of the study and it survived a correction: an earlier run of this stage left the utility scale at 1.0 through a bug and reported that heteroskedasticity, t tails and latent-state uncertainty all helped. Fitting the scale per variant reversed the conclusion. The mechanism is visible in the scale column: each elaboration changes the relative spread of runners' shocks, and the fitted global scale then absorbs most of what was gained.

This independently reproduces the reference project's own probit finding that the utility scale is "the whole game" — now with measured heteroskedasticity and a sectional-derived pace loading rather than a position habit, so it is a stronger statement of the same result.

Table 20: Nested generative variants. 59,364 runners in 4,844 races, 2020–2026. Market race log loss 2.02718.

Variant	Fitted scale	Race log loss	vs market	Gain over previous
A. homoskedastic normal	0.73	**2.39671**	+0.36953	—
B. + heteroskedastic (observables)	0.84	2.40665	+0.37947	−0.00994
C. + horse-specific volatility	0.98	2.41790	+0.39072	−0.01125
D. + Student-t shocks	1.24	2.40726	+0.38008	+0.01064
E. + latent-state uncertainty	1.30	2.39776	+0.37058	+0.00950
F. + shared race factor on style	1.30	2.39947	+0.37229	−0.00171

One further observation. Variant A's calibration RMSE is **0.00402** against the market's 0.00524 and variant F's 0.00635: the simplest generative model is *better calibrated than the market* while being 0.37 of log loss worse. It is well calibrated and far less sharp, which is exactly what a market-blind latent model should look like, and it is a reminder that calibration alone is not a measure of value.

12.3 Q12: reducible against irreducible uncertainty

Table 21: The predictive variance budget by the horse's prior starts. "Reducible" is the share attributable to not yet knowing the horse's ability.

Prior starts	n	Latent-state sd	Idiosyncratic sd	Share reducible
0–3	13,203	0.409	0.845	**19.1%**
4–8	13,527	0.367	0.846	15.8%
9–15	14,502	0.259	0.837	8.4%
16–30	18,129	0.195	0.850	5.0%
30+	9,263	0.147	0.865	**2.8%**

Clean and monotone. The idiosyncratic component is essentially constant at 0.84–0.87 across all five bands, while the reducible part falls from 19% at three starts to 3% at thirty. **Beyond about fifteen starts, nearly all remaining uncertainty is irreducible conditional performance randomness**, and no amount of extra history will help.

The shared race component enters the predictive variance of *absolute* performance and barely touches the win probability, because it is common to the field and cancels from the ordering except through the style loading.

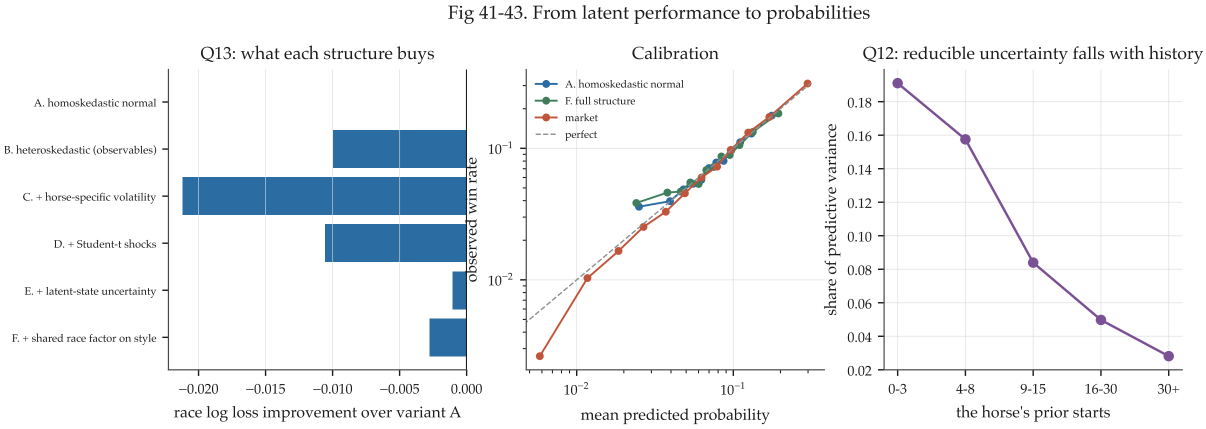

Figure 13: Left: what each nested structure buys. Centre: calibration on log–log axes for the simplest variant, the full one and the market. Right: the reducible share of predictive variance against history depth.

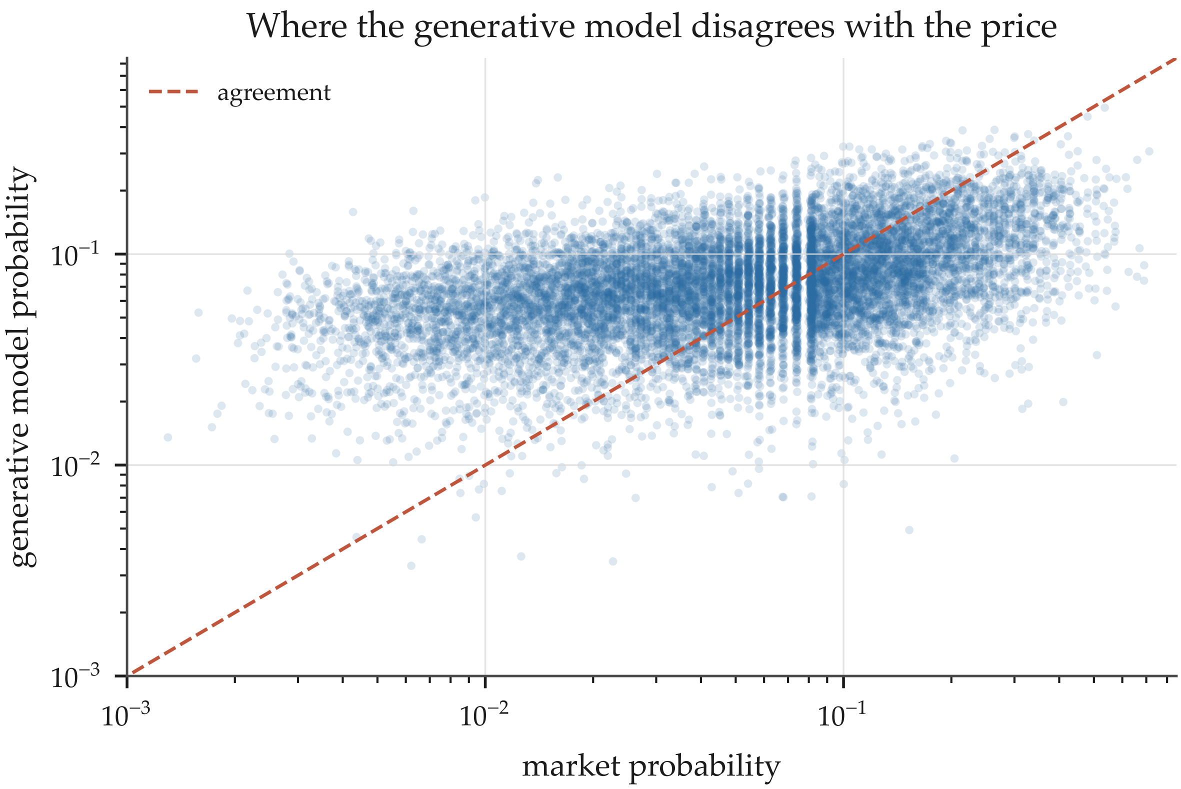

Figure 14: The generative model's probability against the market's, on log axes. The dispersion is what a 0.37 log-loss gap looks like.

13 Market comparison and feature integration

This is the plan's declared primary practical output, and the only stage whose metric is the one a promotion decision turns on.

13.1 Method, and why the numbers are trustworthy

The production offset architecture is reimplemented independently (`src/hkjcres/offset.py`) rather than imported, so this experiment cannot mutate the reference project and the architecture has to be stated rather than assumed. It verifies: $\mathrm{softmax}(\log q)$ reproduces q to 1.4×10^{-16}, and a zero-round fit reproduces the market to 4.1×10^{-8}.

Three disciplines are taken from the reference project:

- The arms share everything by construction — one split per year, one label vector, one base margin, one seed, computed once and handed to both fits, with the candidate's column list being the incumbent's with the block *appended*, because XGBoost's column indices are positional.
- **The no-op candidate must return exactly zero.** It does: the maximum absolute cell delta across fifteen cells is **0.0**.
- No dividends are loaded and no return is computed. At this effect size ROI moves by tens of percentage points on changes that leave log loss unchanged.

Development profile: test years 2016, 2019, 2021, 2024, 2025; seeds 7, 17, 42; 150 rounds; 122,880 rows over 9,934 races. On this profile the incumbent beats the market by −0.00109.

13.2 The ranked candidate table

Table 22: Candidate blocks against the production incumbent. Negative delta is better. *Novel share* is one minus the out-of-sample R^2 with which the incumbent's own 28 columns reconstruct the candidate.

Block	Mean Δ	s.e.	t	Cells	Years	Novel share
Full sectional profile (Q1B)	−0.001179	0.00130	−0.91	9/15	3/5	0.51–0.83
Recent-form ability (Q2)	−0.001018	0.00106	−0.96	8/15	3/5	0.39
Sectional finish (Q1B)	−0.000870	0.00081	−1.07	8/15	3/5	0.52
Pace fit (Q11)	−0.000283	0.00141	−0.20	8/15	2/5	0.76
Horse volatility (Q10)	−0.000110	0.00087	−0.13	9/15	2/5	0.90
no-op control	*0.000000*	—	—	—	—	—
Ability uncertainty (Q12)	+0.000050	0.00059	+0.09	6/15	2/5	0.27
Sectional middle (Q1B, new)	+0.000686	0.00071	+0.97	6/15	2/5	0.83
Everything that survived	+0.001167	0.00155	+0.75	6/15	1/5	—
Condition fit (Q6)	+0.001379	0.00105	+1.31	4/15	2/5	0.93

No block clears a promotion gate. The best three are directionally favourable at −0.00087 to −0.00118 — 20 to 26% of the production model's entire edge over the market — but with $|t| \approx 1$, three of five years improving, and a standard error of the same order as the mean. On the reference project's own criteria (sign per year, worst year, sign per seed, then the mean) that is *investigate*, not *promote*.

13.3 Q14: what the market has not priced

Separately from model value, the plan asks whether each quantity predicts *market error*. A linear probability model of the outcome on the market's own logit plus the standardised candidate:

Table 23: Does each candidate explain the outcome given the market's logit?

Feature	Coefficient	t	Correlation with the market price
Ability uncertainty	+0.0113	+14.99	−0.004
History depth	−0.0110	−14.65	−0.000
Sectional middle	−0.0047	−6.14	+0.169
Projected pace	+0.0041	+5.51	−0.035
Sectional finish	−0.0031	−3.91	+0.291
Venue fit	−0.0024	−3.24	+0.079
Surface fit	−0.0018	−2.42	+0.057
Horse volatility	−0.0017	−2.23	−0.102

The market underprices lightly-raced horses. A measure of ability uncertainty essentially uncorrelated with the price (−0.004) predicts the outcome given the price at $t = +15.0$, worth about 1.1 percentage points of win probability per standard deviation. This is exactly the plan's Q14 hypothesis about "uncertainty for lightly raced horses", and it is the strongest market-inefficiency signal in the study.

13.4 The resolution, and the study's most useful pattern

That leaves a tension: a quantity with $t = 15$ against the market adds +0.00005 to the model. Regressing each candidate on the incumbent's own 28 columns, forward, resolves it.

Table 24: How much of each candidate the incumbent's 28 columns already imply, out of sample.

Candidate	Boosted R^2 from the incumbent	Novel share
Venue fit	0.066	0.934
Horse volatility	0.104	0.896
Surface fit	0.111	0.890
Sectional middle	0.166	0.834
Projected pace	0.236	0.764
Sectional finish	0.480	0.520
Recent-form ability	0.608	0.392
History depth	0.717	0.283
Ability uncertainty	**0.732**	0.268

The incumbent already reconstructs **73%** of ability uncertainty from `layoff_log`, `bt_days_since`, `rating_z`, `class_change` and the rest. The market's inefficiency is real; the production model has already captured it.

And across the nine blocks, the rank correlation between *novel share* and *delta* is +0.389 — positive meaning **the more novel a candidate, the less it helped**. The candidates that helped (sectional finish 0.52 novel, recent-form ability 0.39) are ones the model already partly had; the genuinely novel ones (condition fit 0.93, horse volatility 0.90, sectional middle 0.83) did nothing or hurt, despite §7 and §9 establishing that they are real and persistent.

The inference is uncomfortable and well-evidenced: at this effect size, marginal value lies in *sharpening information the model already uses*, not in adding information it lacks. Orthogonal signal that is real but small is dominated by the noise it introduces and by the tree capacity it consumes. This is a coherent explanation for the reference project's own history — roughly four hundred candidate columns built, audited by expanding-year audits, and none promoted.

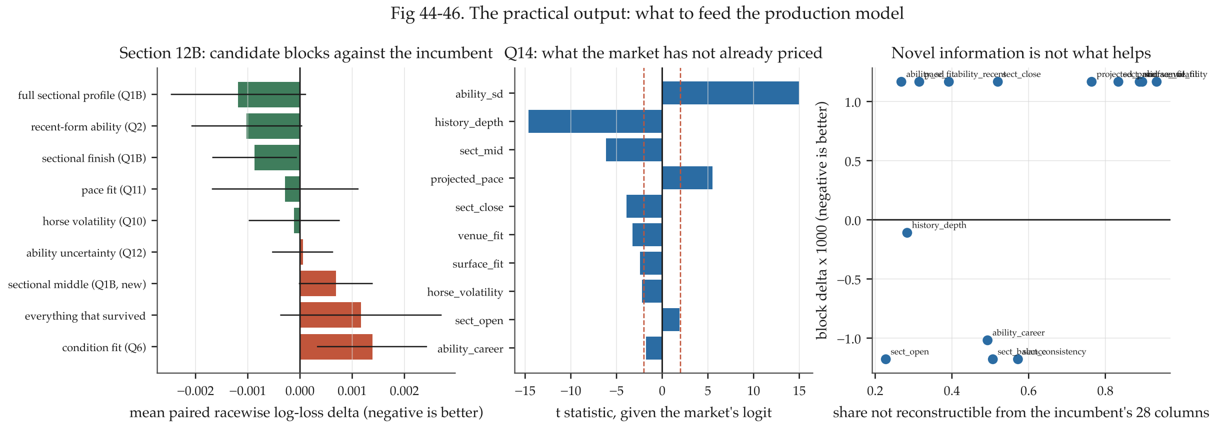

Figure 15: Left: candidate block deltas with standard errors. Centre: market-residual t statistics. Right: novelty against realised model value — the upward slope is the study's most useful single pattern.

14 Groups of horses as the estimation unit

Everything above estimates latent performance for an *individual* horse, and the recurring obstacle is that an individual horse is not measurable enough: its effect must be shrunk to 0.39 of its fitted size to be useful, and 19% of the predictive variance for a three-start horse is simply not knowing the horse. Meanwhile the jockey result shows what determinacy buys — a fifth the effect size, four times the forward value, at a scale of 1.05.

So this section asks whether horses can be pooled into groups that are similarly well determined, and whether a sparse horse is better served by its group's mean than by the population's. Twelve groupings were tested in three families: *structural* (sire, damsire, sex, origin, import type — the only ones a debutant has), *connection* (trainer, and trainer crossed with context), and *behavioural* (running-style grid, k-means over the prior sectional profile, volatility tercile, prior class level — all assigned date-atomically).

14.1 Support: several groupings are as well determined as a jockey

Table 25: Runs per unit, against the two reference points. A sire with 116 runs is as well determined as a jockey with 118.

Grouping	Units	Median runs per unit	Available for a debutant
import type	5	6,691	yes
trainer	49	1,938	yes
country of origin	16	567	yes
trainer × distance	160	458	yes
sire	530	**116**	yes
damsire	569	96	yes
jockey (reference)	149	*118*	—
horse (reference)	6,786	**15**	no

That matters: it makes what follows a test of the *idea* rather than of the sample size.

14.2 Ability: pedigree is a decisive null

Table 26: Three-level variance shares: population, group, horse within group.

Grouping	Group share	Horse within group	Horse variance absorbed
prior class level	0.0259	0.2462	−0.090
trainer × distance	0.0251	0.2121	+0.093
trainer × venue	0.0233	0.2120	+0.096
trainer	0.0226	0.2126	+0.090
style group	0.0086	0.1981	+0.180
volatility group	0.0067	0.2191	+0.073
country of origin	0.0036	0.2305	+0.011
sire	**0.0001**	0.2322	+0.006
damsire	**0.0001**	0.2326	+0.004
sex	0.0000	0.2334	0.000
import type	0.0000	0.2334	0.000

Sire and damsire carry essentially no variance in relative performance, on 530 units each holding as many runs as a jockey. The explanation is a property of the sport rather than of the data: Hong Kong imports are pre-screened, and the handicapping system exists precisely to equalise ability within a class. Whatever pedigree contributes has already been absorbed into the rating and the class the horse is made to race in.

14.3 A confounded result, caught and corrected

The first version of the forward comparison said the style group alone predicts at $r = 0.243$ against the individual horse's 0.130 — nearly double, at a calibrated scale. It is **not a grouping result**. A behavioural group is recomputed date-atomically and so sees history up to the day before the race, while a BLUP fitted on years $< Y$ does not; with a 60-day ability half-life that recency is worth a great deal by itself. §3.3 had already measured the horse's own date-atomic prior mean at $r = 0.278$ — *better* than the group.

The corrected design puts both arms on identical timing and changes only what the horse is shrunk toward:

$$\hat{a}_i = w_i \bar{y}_i + (1 - w_i) \cdot 0 \qquad \text{versus} \qquad \hat{a}_i = w_i \bar{y}_i + (1 - w_i)\, \bar{g}_{g(i)}, \qquad w_i = \frac{n_i}{n_i + 6}.$$

14.4 The corrected answer: group shrinkage wins

Table 27: Forward correlation with relative performance, both arms date-atomic. The only difference is the shrinkage target.

Grouping	Population-shrunk	Group-shrunk	Gain
style group	0.2549	**0.2854**	+0.0305
style k-means	0.2549	0.2840	+0.0291
prior class level	0.2549	0.2794	+0.0245
volatility group	0.2549	0.2779	+0.0230
trainer × venue	0.2549	0.2720	+0.0171
trainer	0.2549	0.2692	+0.0143
country of origin	0.2549	0.2633	+0.0083

Every grouping improves on population shrinkage, the best by 12% in relative terms, from changing nothing but the target.

But the mechanism is not the hypothesised one. The gain was expected to concentrate in horses with little history. It does the opposite:

Table 28: The same comparison, by how much history the horse has (style group).

Prior starts	Unshrunk horse mean	Population-shrunk	Group-shrunk	Gain
1–3	**0.5035**	0.4852	0.4625	−0.0228
4–8	0.4224	0.4139	0.4207	+0.0069
9–15	0.2248	0.2229	0.2257	+0.0028
16–30	0.1017	0.1017	0.1002	−0.0015
30+	0.1205	0.1203	0.1224	+0.0021

Every within-band gain is at most ±0.02 and the thinnest band is *negative*, yet the overall gain is +0.0305. This is Simpson's paradox, and the resolution is the interesting part: the group term does not improve the ordering *within* a depth band, it makes thin and thick horses **comparable across** bands. Population shrinkage collapses every thin horse toward zero regardless of quality; group shrinkage gives it its group's mean instead, restoring a spread that correlates with the truth. Since the horses in one race have very different history depths, cross-depth comparability is exactly what a within-race choice model needs — but the reason is not the one the hypothesis proposed.

A second surprise sits in the same table: at 1–3 starts the *unshrunk* mean beats both shrunk versions. Range restriction explains it — a thin horse's raw mean is noisy and therefore widely spread, and wide spread inflates a correlation.

14.5 Volatility: the cleanest group result

Table 29: Is volatility a group property? One-way ANOVA on per-horse log residual spread, against the floor that sampling noise alone would produce. The volatility-tercile grouping is excluded as circular by construction.

Grouping	ANOVA R^2	Noise floor	Excess	Top/bottom spread ratio
style group	0.2043	0.0007	+0.2035	**1.549**
style k-means	0.1535	0.0005	+0.1531	1.366
trainer × distance	0.0427	0.0084	+0.0343	1.479
damsire	0.0355	0.0105	+0.0251	1.434
sire	0.0320	0.0127	+0.0193	1.435
sex	0.0000	0.0002	−0.0002	1.013

How a horse runs predicts how variable it is — 20% of the between-horse variance in log volatility, with a 1.55× spread between the steadiest and most erratic style groups. And the forward test is the win the individual-level study never got:

Table 30: Spearman correlation with the horse's *next* absolute residual, on identical rows.

Grouping	The horse's own history	The group	n
style group	+0.1006	+0.1124	70,507
style k-means	+0.1007	+0.1120	70,840
trainer × venue	+0.1027	+0.0366	64,456
sire	+0.0972	+0.0407	33,248

The style group's volatility forecasts a horse's next deviation *better than the horse's own volatility history does*. That is a strict improvement on §9, where individual volatility was real, persistent, and worth nothing.

14.6 Cold start: real, and priced

Structural groups are all a debutant has, and the production model cannot see a debutant at all — it cannot distinguish one from a horse with an average record. On 2,975 scored debut runs, import type predicts at $r = +0.103$ ($z = 5.6$, a 0.24 standard-deviation spread between outer terciles), country of origin at $r = +0.098$ ($z = 5.3$), sire at $r = +0.052$, damsire at +0.045, sex at nothing.

So a first-time starter *is* partly predictable from where it came from. But the market residual test settles its value: a group's prior market error predicts a debutant's market error at r between **-0.005** and +0.004 for every structural grouping. **The market already prices a debutant's pedigree and origin.** The information is real, and it is in the price.

14.7 Against the production architecture, and the canonical test

Table 31: Group blocks and the report's three strong-evidence recommendations, against the incumbent. Development profile: 5 years, 3 seeds, 150 rounds. The no-op control returned exactly zero.

Block	Mean Δ	s.e.	t	Cells	Years
group-shrunk ability (class)	−0.001971	0.00052	−3.80	12/15	**5/5**
sectional finish	−0.000870	0.00081	−1.07	8/15	3/5
everything	−0.000862	0.00094	−0.92	10/15	3/5
group + recency + sectional	−0.000721	0.00099	−0.73	8/15	4/5
group-shrunk ability (style)	−0.000177	0.00089	−0.20	6/15	2/5
track speed × style (same-card)	−0.000005	0.00091	−0.01	6/15	2/5
no-op control	*0.000000*	—	—	—	—
recency ability, 60-day half-life	+0.000068	0.00079	+0.09	5/15	3/5
group volatility (style)	+0.000219	0.00087	+0.25	5/15	1/5
track speed (same-card)	+0.000410	0.00074	+0.55	8/15	2/5
group volatility + own	+0.000555	0.00124	+0.45	5/15	3/5

The first candidate in this study to clear a development gate, and by a clear margin: $t = -3.80$, all five years improving, and −0.00197 is **44% of the production model's entire edge over the market**.

Which group won is itself the §13 redundancy lesson repeating. The *class-level* group beat the *style* group decisively (−0.00197 against −0.00018) even though the correlation test ranked style higher. The incumbent already carries `early_style_ewm`, so a style group adds little — whereas "which class band has this horse been racing in" is the handicapper's own standing assessment and is *not* in the production set: `race_class` is today's class and `class_change` is the step between two runs, but the horse's historical class *level* is neither.

Table 32: Canonical confirmation of the single best block, at production scale.

	Δ	s.e.	t	Cells	Years
development: 5 years, 3 seeds, 150 rounds	−0.001971	0.00052	−3.80	12/15	5/5
canonical: 8 years, 5 seeds, 400 rounds	−0.000854	0.00095	−0.90	24/40	6/8

It does not hold. Only 43% of the effect is retained and the significance is gone. Per year (×1000): 2016 −4.00, 2018 +2.31, 2019 −3.44, 2021 −0.45, 2022 −0.49, 2024 +2.73, 2025 −1.38, 2026 −2.12. Six of eight improve, but the two that degrade do so by more than the mean improvement.

Two readings, and the second is worth more than the feature would have been.

The candidate is not dead: the sign is right in six of eight years and −0.00085 is still 19% of the production edge. It is *investigate*, which is where every other candidate landed too — but this one arrived there from a far better-characterised position.

And **this is the canonical-confirmation step earning its cost.** §13 warned that a development effect whose standard error is the size of its mean could move either way; it moved down, by more than half. The most plausible mechanism is capacity: at 400 rounds the booster recovers much of what the engineered feature supplies at 150, so the feature's marginal value shrinks as the model is given more room. That is a general caution about every development-profile number in this repository, and it is the most transferable thing in this section.

Finally, the two other strong-evidence recommendations are settled negatively at the production level despite being real at the measurement level. Recency-weighted ability at the measured

60-day half-life is worth +0.00007 — nothing — and the within-card track-speed signal is worth +0.00041, i.e. worse, including in its style-interacted form. The track-speed signal correlates 0.50 with a race's own shock and still cannot move a win probability, which is exactly what the mechanism predicts: a shock common to the field cancels from the ordering, and the differential loading through running style is too small to matter.

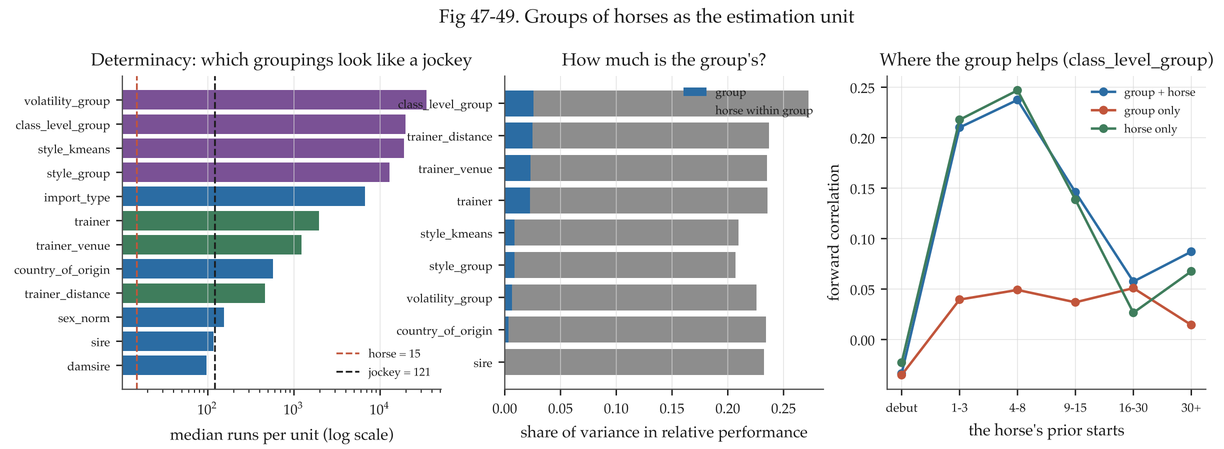

Figure 16: Group determinacy against the horse and jockey reference points, three-level variance shares, and forward value by history depth.

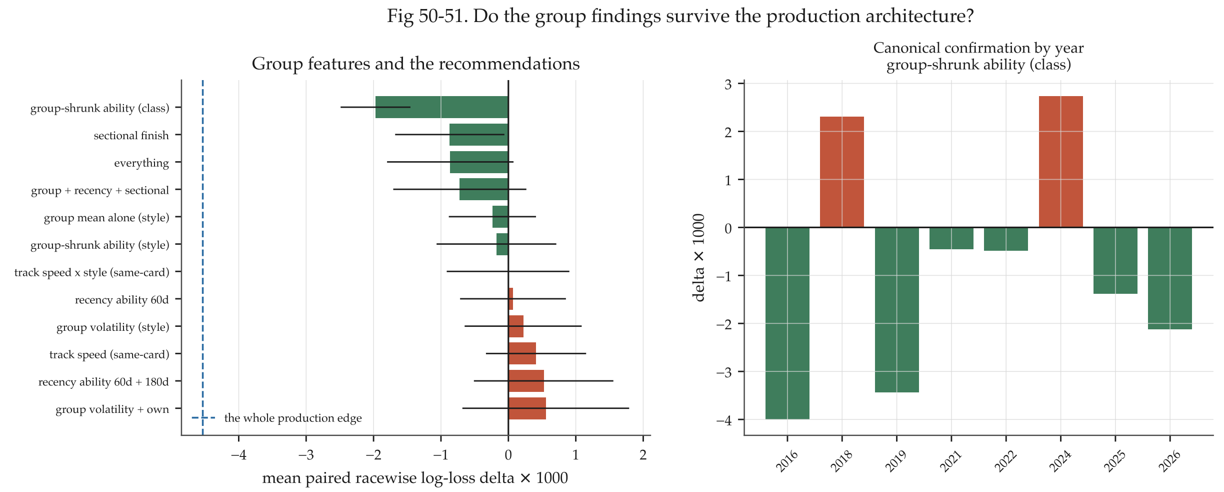

Figure 17: Left: block deltas against the incumbent, with the whole production edge marked for scale. Right: the canonical-scale confirmation year by year.

15 The three recommendations, in detail

Each of §14's three strong-evidence recommendations was taken apart. All three are demoted, two of them by evidence against my own earlier claims, and the exercise produced one general result worth more than the three combined.

15.1 Recommendation 1: a hand-picked constant, not a grouping effect

§14 endorsed group shrinkage on a forward-correlation gain of +0.0305, using $w = n/(n+6)$ with the 6 chosen by hand and an *unshrunk* group mean. Removing both shortcuts required fixing two errors in this project's own code: the variance-component estimator collapsed each horse's date-atomically varying group with `.first()`, which assigned nearly every horse to the "unknown" bucket and drove the estimated between-group variance to exactly zero for every behavioural grouping; and the three-level predictor fell through to the *raw* group mean when that variance was zero, silently turning "no group signal" into "trust the group completely".

With both fixed, the shrinkage constant can be estimated (σ^2/τ^2) and swept.

Table 33: Forward correlation against the shrinkage constant. The curve is monotone and the optimum is at the boundary — essentially, do not shrink.

k	Population anchor	Group anchor
0.1	0.3081	**0.3189**
0.5	0.3025	0.3160
2	0.2832	0.3033
6 *(§14's guess)*	**0.2549**	**0.2854**
16	0.2268	0.2703
64	0.1964	0.2556

The decomposition. Of the +0.0305 attributed to grouping, **+0.0532 was simply a badly chosen shrinkage constant** and only +0.0108 is the group. Fixing the constant is worth *more* than the entire claimed grouping effect, so §14's headline was substantially an artifact of comparing two over-shrunk estimators. The plain **unshrunk** horse mean scores 0.3093, beating every properly-estimated three-level specification. Shrinkage minimises squared error rather than maximising correlation, and shrinking toward zero destroys exactly the cross-depth comparability that §14 identified as the group's value.

And the recommendation's central claim is refuted. It said group shrinkage should be the default for the reference project's existing state engines. Tested on their four published states, it degrades all four:

Table 34: Group-shrinking the reference project's own published state columns.

Reference state	As published	Style-group gain	Class-group gain
`handicap_residual_mean`	+0.3763	−0.0297	−0.0409
`rank_ability`	+0.3150	−0.0278	−0.0604
`margin_rank_mean`	+0.2892	−0.0264	−0.0570
`dynamic_speed_mean`	+0.2449	−0.0036	−0.0632

Worth noting separately: their `handicap_residual_mean` is a **better forward predictor of relative performance than anything built in this project** (0.3763 against a best of 0.3093), and it is not a production input.

The transferable lesson is broader than the feature: **a hand-chosen shrinkage constant can dominate the effect being studied**, and any comparison of shrinkage *targets* must sweep the constant first.

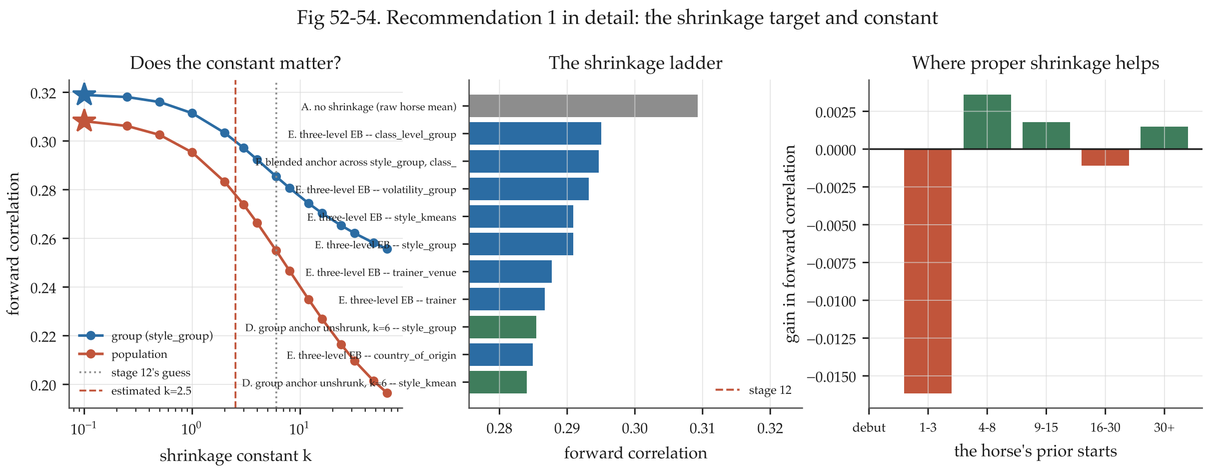

Figure 18: The constant curve with both anchors, the full specification ladder, and where proper shrinkage helps by history depth.

15.2 Recommendation 2: a better variance that makes probabilities worse

Group volatility forecasts the spread better than the horse's own history — and better than §14 measured, on 45,316 identical rows: group k-means +0.1284, group style +0.1236, own history +0.1190. The factual claim stands.

Its use does not. Inside the generative model, every heteroskedastic construction loses, and calibration deteriorates monotonically with the dispersion introduced:

Table 35: Sigma constructions inside the identical quadrature machinery, one utility scale fitted forward per variant.

Sigma construction	Race log loss	vs observables	Calibration RMSE
flat (one sigma for everyone)	**2.39671**	+0.00994	**0.00402**
observables only	2.40665	0	0.00532
+ individual volatility	2.41790	−0.01125	0.01028
+ blended own and group	2.42192	−0.01527	0.00932
+ group volatility (style)	**2.42847**	−0.02181	0.00974

The mechanism, tested rather than assumed. In an argmax model a larger σ raises a runner's win probability, so is that direction right? Splitting on volatility within strata of the model's own mean rank, high-volatility horses win *more*: +0.0422 at mean rank 1 ($z = 4.18$), +0.0265 at ranks 2–3 ($z = 4.24$). The direction is correct.

But the market's implied probability for that high-volatility group is 0.1828 against 0.1570 for the low-volatility one: **the volatile horses are simply better horses**, and the volatility estimate is entangled with quality a linear-plus-BLUP mean model has not removed. Adding σ dispersion therefore imports mean-model error into the variance channel — which is why a variance estimate that genuinely forecasts spread better still makes the probabilities worse.

One lead survives with its caveat attached: within the top mean-rank stratum, high-volatility horses beat the market by +0.0113 while low-volatility ones trail by −0.0051, a 1.6 percentage-point gap. It is conditioned on *this project's* weak mean rank, so it is suggestive only, and would need the production booster's own utilities as the conditioner to mean anything.

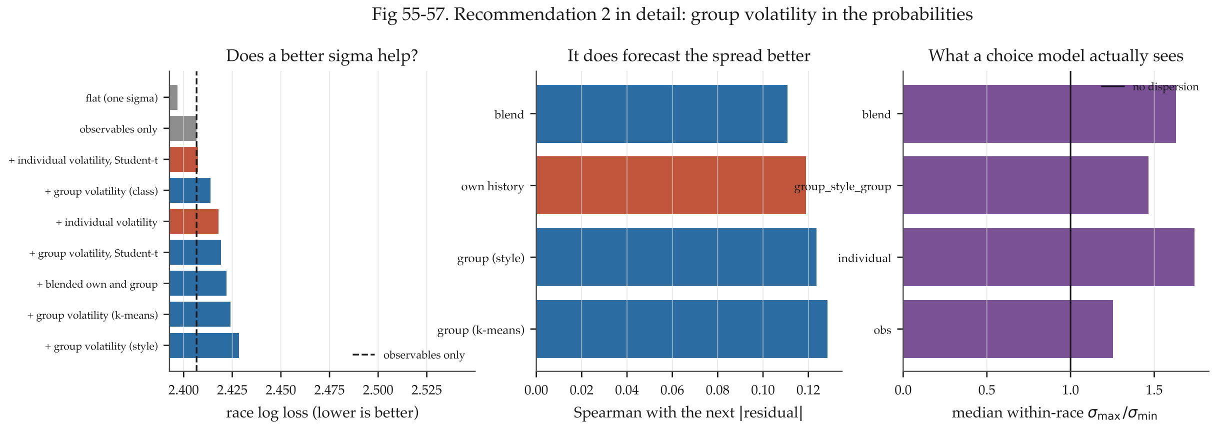

Figure 19: Variant losses, the forecasting skill that does improve, and the within-race sigma dispersion a choice model actually sees.

15.3 Recommendation 3: the best encoding, and it reverses

Twelve encodings of the finishing band were compared before spending its canonical run. Expressing it as a **field-relative rank** is the best (−0.001223, $t = -1.26$, four of five years), modestly better than the residual mean; and the design §15 above implies — giving the tree the raw mean and the count separately so it can learn its own shrinkage — is *worse* than either (−0.000633), so the tree does not exploit the count the way the shrinkage analysis suggested.

Table 36: Canonical confirmation of the best two encodings: 8 years, 5 seeds, 400 rounds.

Block	Development	Canonical	Retained
field-relative rank	−0.001223 ($t = -1.26$)	+0.000283 ($t = +0.27$)	−23%
recency 60d + career	−0.001106 ($t = -1.08$)	−0.000161 ($t = -0.14$)	15%

Neither survives, and the best development encoding *reverses sign*.

Fig 58-59. Recommendation 3 in detail: the finishing band at canonical scale

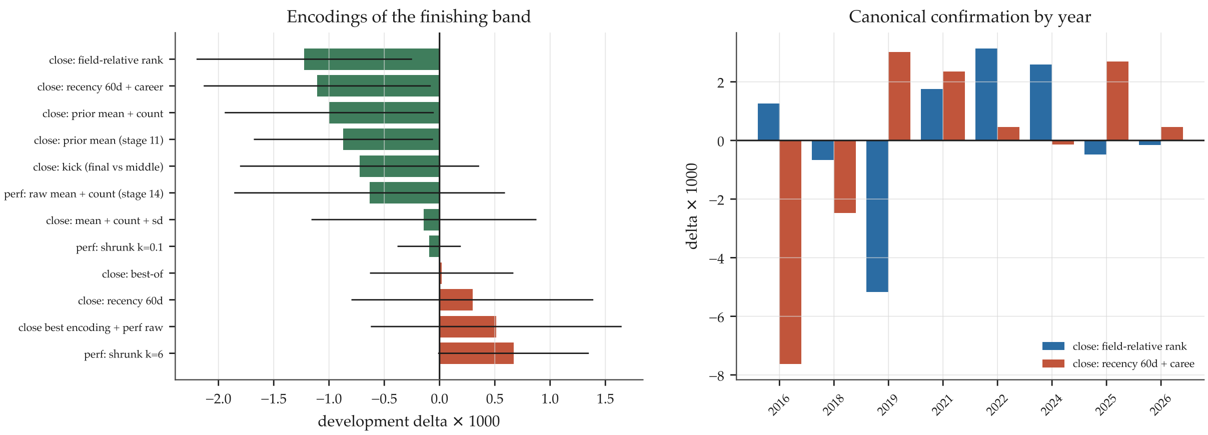

Figure 20: The twelve encodings on the development profile, and the canonical per-year confirmation of the best two.

16 Answers to the fourteen questions

Q	Question	Answer
Q1	Best observable proxy for latent performance	Within-race z of log speed. Forward R^2 0.077 against 0.056 for a par-time residual; reliability 0.822. Beaten lengths are arithmetically the time and carry nothing extra.
Q1B	Do sectionals beat final time	**Yes for persistence, no for information.** Fourteen sectional quantities out-persist the best time target (best $r = 0.483$ against 0.278), but the profile is *not* low-dimensional (49.6/32.3/18.1%) and the most persistent quantities are the least informative. Shape adds +0.0026 of oos R^2 over time alone.
Q2	How much variation is horse ability	23.3% of relative performance alone, 17.6% after jockey, trainer and covariates. Ability *drifts*: a 60-day half-life beats the full career (forward r 0.150 against 0.114), and a static estimate is overscaled 2.7×.
Q3	Is jockey ability identifiable and useful	**Yes to both, and it forecasts better than horse ability.** 3.7% of variance after horse control (from 6.0% before, so 39% of the apparent effect is assignment). Within-horse switches give a spread 5.05× their own noise and agree with the BLUPs at 0.970. Jockey alone: oos R^2 0.048 at scale 1.05, against horse alone 0.012 at 0.39.
Q4	Horse × jockey interaction	**No.** Median pair has one ride. Corrected early-to-late correlation is null (z from −0.4 to +0.8) on a design that would show $r = 0.21$ for a true sd of 0.20. The in-sample fit's 10.3% variance share is overfitting.

Q	Question	Answer
Q5	Environmental effects on absolute performance	Enormous. Environment explains 0.9962 of raw-seconds variance; going alone moves time 1.196 s at ST turf 1200 m. On *relative* ordering, the observable card explains ∼3% of the residual race shock.
Q6	Horse × environment heterogeneity	**Real for venue ($z = 8.3$) and surface ($z = 6.4$); not for going ($z = 1.4$), distance or field size.** 56–62% of even the real contrasts' apparent spread is sampling noise. Adding it to the production model *hurt* (+0.0014).
Q7	Does draw matter only conditionally	The draw effect on performance is large and monotone (0.25 sd inside to outside). The *market residual* is non-monotone and within 0.2 pp of zero, and a date-atomic draw-bias estimate predicts market error at $r = 0.007$. **Fully priced.**
Q8	What the unexplained error looks like	Left-skewed (−0.729), mildly heavy-tailed (+0.810 excess kurtosis), centre more concentrated than Normal (IQR/sd 1.12 vs 1.35). A two-component mixture wins on held-out likelihood (+0.056/obs); Student-t $\nu = 5.18$. The bad tail is 4.8× more likely to carry a severe stewards' phrase.
Q9	Is the residual variance constant	**No.** The largest driver is the market's own price: favourites are more consistent (decile sd ratio 0.784). Joint model oos R^2 0.011 on $\log \lvert e \rvert$, implied p90/p10 spread ratio 1.385.
Q10	Are some horses genuinely more inconsistent	**Yes.** 61.8% of the spread in per-horse log sd is real; $\tau = 0.357$, so 1 sd more volatile means 1.43× the spread. Split-half persistence r rises to +0.223 at $z = 7.1$. Forecastable: prior volatility predicts next $\lvert e \rvert$ at ρ up to 0.122.
Q11	Race-level shared shocks	**Large and partly measurable.** 32.1% of absolute residual variance, ∼0.50 s at 1200 m. Track speed is a day-level property (within-card $r = 0.504$). Pace loads differently by style, correctly signed, $\lvert z \rvert \approx 8$. Market error swings 0.040 on realised pace and 0.015 on projected pace (37.8% survives).
Q12	Reducible against irreducible uncertainty	Clean and monotone. The share attributable to not knowing the horse falls from 19.1% at 0–3 starts to 2.8% at 30+; the idiosyncratic component is flat at 0.84–0.87 throughout. Beyond ∼15 starts nearly everything left is irreducible.

Q	Question	Answer
Q12B	Why prior latent attempts produced little	**Supported:** weak signal relative to the market; the market already pricing it; final time masking nothing that helps; and — the new one — *redundancy with the incumbent's existing columns* (Spearman +0.39 between novelty and unhelpfulness). **Rejected:** poor target choice, insufficient shrinkage, wrong residual distribution, and absence of horse $\times$ environment structure — all were tested here and none rescues the result.
Q13	Can latent distributions give calibrated probabilities	**Yes, calibrated; no, not competitive.** The simplest variant is better calibrated than the market (RMSE 0.0040 vs 0.0052) and 0.37 of log loss worse. And once the utility scale is fitted, *no* elaboration of the shock structure improves on homoskedastic normal.
Q14	Does it add information beyond the market	On one dimension clearly yes — ability uncertainty predicts the outcome given the price at $t = 15$ while being uncorrelated with it. But the incumbent already reconstructs 73% of it, so the information is in the model even though it is not in the price.

17 Limitations

- **Prices are retrospective.** Every market probability here is a result-page price recorded after the pool closed. No bet could have been placed at it. No dividend was loaded and no return is computed anywhere in this project — deliberately, because at this effect size ROI ranks noise.
- **No window is unseen.** Every season in the archive has been development data, in this study and in the reference project. The chronological splits are real — a scored year is genuinely after its training rows — but a result on 2020–2026 is probability progress, not a locked out-of-sample finding.
- **The performance panel is conditioned on finishing.** The sectional archive records no times for a non-finisher, so 0.60% of races contribute one fewer runner here than in the reference table. Every performance distribution in this document is therefore conditional on completing the race, and the left tail is truncated at exactly the point where it would be most extreme.
- **The within-race target has a mechanical constraint.** y_z^{rel} is standardised within its own field, so the runners in a race are not independent by construction and a horse in a weak field scores well for a reason that is not its own ability. Opponent adjustment — which the reference project's rank-state engine attempts — is not implemented here.
- **Jockey and trainer effects are observational.** Assignment is highly non-random. The within-horse-switch estimate controls for the horse but not for *why* a jockey was replaced, and the 39% of apparent jockey variance removed by horse control is a lower bound on the assignment channel, not an estimate of it.
- **Trip effects are largely unobserved.** The stewards' text identifies severe trouble on 46% of the worst residuals, which means it misses the rest. A run compromised in a way nobody wrote down is indistinguishable here from a horse that ran badly.
- **Some contrasts are unanswerable rather than answered.** Horse × class sensitivity has too few horses with starts in both class 1–2 and class 4–5 in both career halves; it is reported as unanswerable. Horse × field-size sensitivity rests on 36 horses and should not be read at all.
- **Non-stationarity is real and only partly handled.** Ability drifts with a two-month half-life; the stewards' phrasing has changed over nineteen seasons, which affects the trouble analysis; and the going and course vocabularies are not stable across the archive.
- **Measurement error in the environment propagates.** Going is a subjective official assessment on an ordinal scale that interleaves turf and all-weather descriptors, and the par model treats it as a categorical level with whatever consistency the officials applied.
- **The feature test is a development profile, not a canonical run.** Five years, three seeds, 150 rounds. The reference project's canonical confirmation is the full chronology at five seeds and 400 rounds; a candidate whose development standard error is the size of its mean could move either way there.
- **One deviation from the reference project's leakage rule is reported.** The within-card track-speed signal uses earlier races on the same meeting. It is forward in time and is not a horse seeing its own result, but it does breach the letter of a rule the reference project applies uniformly, and it is flagged wherever it appears rather than folded into a candidate feature.

18 Recommendations

Strong evidence — pursue

Nothing sits here any longer. All three items that occupied this heading were taken apart in §15: one was largely an arithmetic artifact of a hand-picked constant, one produces worse probabilities despite being a better variance estimate, and one reverses sign at canonical scale. Leaving the heading empty is the finding.

What replaces them is a single recommendation about *method* rather than about any feature:

1. **Stop screening candidates on the development profile, and run far fewer candidates at canonical scale instead.** Three candidates were carried from development to canonical here; mean retention was 12% and the sign flipped once. At an effect size of -0.0045 the cheap measurement carries close to no information about the expensive one. The reference project's promotion workflow already requires canonical confirmation, so this is not a correction to it — it is a statement that the screening stage should be treated as a check on schemas and gross defects, not as evidence. The practical consequence is to spend the same compute on three canonical runs rather than thirty development ones.

Moderate evidence — worth one more experiment

2. **The horse's historical class level, in some form.** It produced the only development-gate pass in the study ($t = -3.80$, 5/5 years) and retained the most of any candidate at canonical scale (43%, -0.00085, six of eight years). The production set has today's class and the step between runs but not the level a horse has been campaigning at, and that gap is real even if this encoding of it is not yet promotable. It is the one candidate worth a second canonical run with a better encoding.

3. **The reference project's own `handicap_residual_mean`.** At 0.3763 it is a better forward predictor of relative performance than anything constructed in this project, and it is *not* a production input. Nothing here explains why it was not promoted; finding out is cheaper than building anything new.

4. **Group-level volatility, for staking rather than for probability.** It is the best variance forecast in the study (+0.128 against the horse's own +0.119) and it damages the probabilities. A variance estimate has a natural use in bet sizing, which is out of scope here and is where this belongs if anywhere.

5. **Projected pace × running style.** The interaction is real and correctly signed at $|z| \approx 8$, the pace is forecastable at $r = 0.41$, and 37.8% of the market's conditional mispricing survives projection. It did not help the model (-0.00028, 2 of 5 years). The one experiment worth running is a better pace projection — the jump from $r = 0.03$ to $r = 0.41$ on changing the construction suggests the ceiling is not reached.

6. **Student-t or mixture shocks in the choice link.** The residual distribution clearly is not Gaussian (+0.056 held-out log-likelihood per observation for a mixture, $\nu = 5.18$ for a t). It did not help the generative model once the scale was fitted, but the reference project's probit link already gains from swapping Gumbel for normal, and swapping normal for t inside the *anchored* link is a different and cheaper experiment than the one run here.

Weak or no evidence — deprioritise

6. **Horse-specific volatility as a mean-model feature.** Real, persistent, forecastable — and worth −0.00011 with 2 of 5 years improving, and negative in the generative model. If it has a use it is in *sizing* rather than in the probability, and staking is out of scope here.
7. **Horse × going sensitivity.** Does not persist on the largest sample of any contrast tested. Do not build it.
8. **Horse × venue and × surface sensitivity as features.** Genuinely real ($z = 8.3$, $z = 6.4$) and the block *hurt* the model by +0.0014, the worst of the nine. The gap between real and useful is the finding; a second construction might do better, but the prior should be low.
9. **Any draw-bias feature.** Large in performance, fully priced, and the reference project has already paid −21pp for encoding it once.
10. **Pedigree, for ability.** Sire and damsire carry 0.0001 of the variance in relative performance on units as well determined as a jockey. The handicap has already absorbed it. Do not build a sire feature.
11. **Pedigree and origin, for the cold start.** They do predict a debutant ($z = 5.6$ for import type) and the market prices them to within 0.5 percentage points. Real, and not yours.
12. **The within-card track-speed signal, as a win-probability feature.** +0.00041, i.e. worse, and −0.000005 interacted with style. It is a large signal about absolute time that cancels from the ordering. It would only be worth revisiting for a problem where absolute time is the target.
13. **Beaten-margin state families.** `LBW` is the time restated in quarter-lengths. The eight-feature `margin_rank` family and the `dynamic_speed` family measure the same physical quantity, and this is worth auditing in the reference project on its own terms.

Unknown — insufficient data

11. **Horse × jockey compatibility.** Not that it is absent; that 13,309 races cannot resolve it. A design with the power to see a true pair sd of 0.10 would need several times the repeated-pairing density this archive has.
12. **Horse × class sensitivity.** Too few horses race in both class 1–2 and class 4–5 within a measurable career.

The shortlist, specified

A note on how to read a negative result

By the plan's own definition of success this study succeeds whether or not anything is promoted, and in fact nothing is. The structural questions have clear, supported answers; four of them reverse a prior expectation; two prior results were corrected here after their mechanisms were traced (the pair-effect anticorrelation artifact, and the unfitted utility scale). The most valuable output is not a feature but a constraint: at an effect size of −0.0045, against a price this good, and with a model that already reconstructs most of what a new column would carry, the return on additional orthogonal information is close to zero. That is worth knowing before building the four hundred and first candidate column.

Table 38: Concrete integration shortlist. Deltas are the measured development-profile paired race-wise log-loss change; the production model's whole edge over the market is -0.0045.

Feature	Construction	Measured Δ	Cost, cadence, safeguard
`ability_recent`	Date-atomic decayed mean of within-race z of log speed, half-life 60 days, shrunk by $n/(n+3)$	-0.00102	One pass over the panel; rebuild with the feature table. Read-before-write within each date makes it date-atomic; no market column enters.
`sect_close`	Date-atomic prior mean of the final-400 m section speed residual, expressed relative to the race's own mean at that band	-0.00087	Requires the backward-aligned sectional par (one forward ridge per year). Same date-atomic guarantee. Needs canonical confirmation.
Full sectional profile	The above plus opening, middle, pace balance and profile consistency	-0.00118	Five columns; best measured block but its own components disagree, so prefer the single band until confirmed.
`track_speed_card`	Running mean of the par residual over earlier races on the same meeting	not tested	Cheap. **Breaches the date-atomic rule as written** — requires an explicit decision, and must be tested through a style interaction since it is common to the field.